Advanced Calculus

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1 Metric Spaces

Definition 1.1. Let X be a set, a *metric* d is a map

$$d\colon X \times X \to \mathbb{R}$$

such that:

(i) $d(x, y) \geq 0$ with equality iff $x = y$.

(ii) $d(x, y) = d(y, x)$

(iii) $d(x, z) \leq d(x, y) + d(y, z)$

Example 1.2. Given a vector space V and norm $\|.\|$ we can define the metric

$$d(x, y) = \|x - y\|$$

Question. Given a vector space V and a metric d on V, does d "come from" a norm? i.e does there exist some norm $\|\|$ s.t

$$d(x, y) = \|x - y\|$$

The answer in general is no (consider the discrete metric, or even any bounded metric: this contradicts homogeneity).

Notation. Take (X, d) metric space $\delta > 0$ and $x_0 \in X$, the *δ-neighborhood* of x_0

$$N_\delta(x_0) := \{x \in X : d(x, x_0) < \delta\}$$

Definition 1.3. (X, d) metric space and $E \subset X$

(i) $x_0 \in X$ is an *interior point* of E iff $\exists \delta > 0$ s.t

$$N_\delta(x_0) \subset E$$

(ii) $x_0 \in X$ is a *limit point* iff $\forall \delta > 0$

$$N^*_\delta(x_0) \cap E \neq \varnothing$$

(We denote by E' the set of limit points of E)

(iii) E is *open* iff every $x_0 \in E$ is an interior point of E.

(iv) E is *closed* iff every $E' \subset E$.

(v) The *closure* of E denoted $\overline{E}$ is

$$\overline{E} := E \cup E'$$

Review. Fix (X, d) a metric space.

1. $N_\delta(x_0)$ is open $\forall \delta > 0,\ \forall x_0 \in X$.
2. E is open iff E^c is closed.
3. $\overline{E}$ is closed, in fact $\overline{E}$ is the smallest closed set containing E.
4. The union of any collection of open sets is open.
5. A finite intersection of a collection of open sets is open.
6. Intersection of any collection of closed sets is closed.
7. Finite union of closed is closed.
8. If $x \in E'$ then $N_\delta(x) \cap E'$ is an infinite set.

Definition 1.4. Let x_n be a sequence in X, we say that x_n converges to $x_0 \in X$ iff $\forall \varepsilon > 0\ \exists N \in \mathbb{N}$ s.t

$$d(x_n, x_0) < \varepsilon \quad \forall n \geq N$$

1. The limit of a sequence (if exists) is unique.
2. x_n converges to x_0 in (X, d) iff $d(x_n, x_0)$ converges 0 in $(\mathbb{R}, |||)$.

Example 1.5. $x_n = (1/n + 1, e^{-n})$ in $\mathbb{R}^2$. Find the limit of x_n in $(\mathbb{R}^2, \|.\|_\infty)$, $(\mathbb{R}^2, \|.\|_1)$, $(\mathbb{R}^2, \|.\|_2)$.

- $$\|(1/n + 1, e^{-n}, (1, 0))\|_\infty = \max(1/n, e^{-n}) \leq 1/n + e^{-n} \to 0$$
- $x_0 = (1, 0)$
$$\|(1/n + 1, e^{-n}, -(1, 0))\|_1 = \|(1/n, e^{-n})\|_1 = |1/n| + |e^{-n}| \to 0$$
- $$\|(1/n + 1, e^{-n}) - (1, 0)\|_2 = \sqrt{1/n^2 + e^{-2n}} \to 0$$

Remark 1.6. $x_n \to x_0$ in $(\mathbb{R}^k, \|.\|_\infty)$

$$(x_{1,n}, x_{2,n}, \cdots, x_{k,n}) \to (x1, 0, \cdots, x_{k,0})$$

Iff

$$\begin{aligned} x_{1,n} &\to x_{1,0} \\ x_{2,n} &\to x_{2,0} \\ &\vdots \\ x_{k,n} &\to x_{k,0} \end{aligned}$$

in $(\mathbb{R}, \|.\|)$.

Proof. $|x_{i,n} - x_{i,0} \leq \|x_n - x_0\|_\infty \leq |x_{1,n} - x_{1,0}| + \cdots + |x_{k,n} - x_{k,0}|$ Therefore $x_n \to x_0$ in $(\mathbb{R}^k, \|.\|_\infty)$ $\iff$ $\|x_n - x_0\|_\infty \to 0$ and $\iff$ $|x_{i,n} - x_{i,0}| \to 0\ \forall i$ and hence $\iff$ $x_{i,n} \to x_{i,0}\ \forall i$. □

Remark 1.7. Take $x \in \mathbb{R}^n$, then

$$\|x\|_\infty = \max_i(x_i) \leq |x_1| + \cdots + |x_n| = \|x\|_1 \leq n\|x\|_\infty$$

Therefore

$$\|x\|_\infty \leq \|x\|_1 \leq n \cdot \|x\|_\infty$$

Similarly, show something for $\|x\|_2$ and $\|x\|_\infty$.

Definition 1.8. Given a vector space V and norms $\|.\|_a$ and $\|.\|_b$ we say that $\|.\|_a$ and $\|.\|_b$ are equivalent iff $\exists c_1, c_2 > 0 \in \mathbb{R}$ s.t $\forall x \in V$

$$c_1\|x\|_b \leq \|.\|_a \leq c_2\|x\|_b$$

Lemma 1.9. *(X, d) metric space and $E \subseteq X$, then E is closed iff for sequence $x_n \in E$ that converges $\lim_{n\to\infty} x_n \in E$.*

Proof. $\implies$ Assume that there exists a sequence x_n in E that converges to x_0 s.t $x_0 \notin E$, take $\varepsilon > 0$, then since $x_n \to x_0$ then $\exists N \in \mathbb{N}$ s.t

$$x_n \in N_\varepsilon(x_0) \quad \forall n \geq N$$

But $x_n \in E$,

$$N^*_\varepsilon(x_0) \cap E \neq \varnothing$$

Therefore

$$x_0 \in E'$$

Contradiction to the fact that E is closed.

- Take $x_0 \in E'$, let $\varepsilon_n = 1/n$ then

$$\exists x_n \in N^*_n(x_0) \cap E$$

Consider $\{x_n\}_{n\in\mathbb{N}}$ x_n is a sequence in E s.t

$$d(x_n, x_0) < 1/n \implies x_n \to x_0$$

and hence $x_0 \in E$.

□

Theorem 1.10. *Given a vector space V and norms $\|.\|_a$ and $\|.\|_b$ then the following are all equivalent:*

1. *$\|.\|_a$ and $\|.\|_b$ are equivalent ($c_1\|x\|_b \leq \|x\|_a \leq c_2\|x\|_b$).*
2. *$x_n \to x_0$ in $\|.\|_a \iff x_n \to x_0$ in $\|.\|_b$.*
3. *E is closed in $(V, \|.\|_a) \iff E$ is closed in $(V, \|.\|_b)$.*
4. *U is open in $(V, \|.\|_a) \iff U$ is open in $(V, \|.\|_b)$.*

Proof. $1 \implies 2$

$$c_1\|x_n - x_0\|_b \leq \|x_n - x_0\|_a \leq c_2\|x_n - x_0\| \leq \frac{c_2}{c_1}\|x_n - x_0\|_a$$

Apply Squeeze.

$2 \implies 3$ By the Lemma.

$3 \implies 4$ Trivial.

$4 \implies 1$ $B_{\|.\|_a}(0, 1)$ open in $(V, \|.\|_a)$ therefore it is open in $(V, \|.\|_b)$. Hence 0 is an interior point in $(V, \|.\|_b)$. Hence $\exists \delta > 0$

$$B_{\|.\|_b}(0, \delta) \subseteq B_{\|.\|_a}(0, 1)$$

Let $x \in V$, $x \neq 0$.

$$\frac{\delta}{2} \cdot \frac{x}{\|x\|_b} \in B_{\|.\|_b}(0, \delta) \subseteq B_{\|.\|_a}(0, 1)$$

Therefore

$$\|\frac{\delta}{a}\frac{x}{\|x\|_b}\|_a < 1$$

We get

$$\frac{\delta}{2\|x\|_b}\|x_a\| < 1 \implies \|x\|_a \leq \frac{2}{\delta}\|x\|_b \; \forall x \in V$$

□

1.1 Compact Sets

Definition 1.11. (X, d) metric space, $K \subseteq X$. We say that K is *compact* iff for every open cover

$$\mathcal{G} = \{G_\alpha\}_{\alpha \in I}$$

of K has a finite subcover.

Example 1.12. $(0, 1]$ is not compact in $(R, |.|)$, take

$$\mathcal{G} = \{(1/n, 10)\}_{n \in \mathbb{N}}$$

which has no finite subcover.

Theorem 1.13 (Heine-Borel)**.** *In* $\mathbb{R}$ $[a, b]$ *is compact.*

Example 1.14. If X is finite, then X is compact.

Proof. Let $X = \{x_1, \cdots, x_n\}$ and let

$$\mathcal{G} = \{G_\alpha\}_{\alpha \in I}$$

be an open cover X. Then for every x_i, fix some G_{α_i} s.t

$$x = \{x_1, \cdots, x_n\} \subseteq \bigcup_{i=1}^{n} G_{\alpha_i}$$

□

Proposition 1.15. (X, d) *metric space and* $K \subseteq X$ *then every closed subset* E *of* K *is compact.*

Proof. Let $\mathcal{G} = \{G_\alpha\}_{\alpha \in I}$ be an open cover of E, then

$$E \subseteq \bigcup_{\alpha \in I} G_\alpha$$

Therefore

$$K \subseteq \bigcup G_\alpha \cup E^c, \quad E^c \text{ open}$$

Therefore

$$\mathcal{G}' = \mathcal{G} \cup \{E^c\} \quad \text{is an open cover of } K$$

Therefore it has a finite subcover and

$$K \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_k} \cup E^c$$

Hence

$$E \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_k}$$

Which is a finite subcover of $\mathcal{G}$. □

Definition 1.16. (X, d) metric space and x_n sequence in X and

$$\varphi \colon \mathbb{N} \to \mathbb{N} \quad \text{strictly increasing}$$

We say that $b_n = a_{\varphi(n)}$ is a *subsequence* of a_n.

Proposition 1.17. *x_n converges to $x_0 \iff$ every subsequence x_{n_k} converges to x_0.*

Definition 1.18. (X, d) metric space and $K \subseteq X$. We say that K is sequentially compact iff every sequence $x_n \in K$ has a convergent subsequence in K.

Proposition 1.19. *(X, d) metric space, K sequentially compact iff every infinite $E \subseteq K$ has a limit point in K.*

Proof. $\implies$ Let K be sequentially compact, take $E \subseteq K$ an infinite set. We can extract from E a sequence of distinct elements $x_n \in E \subseteq K$. Therefore x_n has a convergence subsequence x_{n_k} s.t the limit of $x_{n_k} = x_0 \in K$. Moreover, x_{n_k} are distinct in E therefore $x_0 \in E' \cap K$.

$\impliedby$ Let x_n be a sequence in K, take

$$E = \{x_n\}_{n \in \mathbb{N}}$$

We have two cases:

- E **is finite.** Then $\exists x_0$ and $\exists n_1 < n_2 < \cdots$ (infinitely many) s.t

$$x_{n_k} = x_0, \quad \forall k \in \mathbb{N}$$

  Therefore

$$\lim_{k \to \infty} x_{n_k} = x_0 \in E$$

  We found a converging subsequence.
- E **is infinite.** Then by the assumption, we have that E has a limit point x_0 in K.

  $\varepsilon = 1$ $\exists n_1$ s.t $x_{n_1} \in N_1^*(x_0) \cap E$.

  $\varepsilon = 1/2$ $\exists n_2 > n_1$ s.t $x_{n_2} \in N_{1/2}^*(x_0) \cap E$ (since $N_{1/2}^*(x_0) \cap E$ is infinite).

  we can construct $n_k > n_{k-1}$ s.t $x_{n_k} \in N_{1/k}^*(x_0) \cap E$, therefore

$$\lim_{n \to \infty} x_{n_k} = x_0 \in K$$

□

Example 1.20. In $(\mathbb{Q}, |.|)$, $K = \mathbb{Q} \cap [0, 1]$ is closed and bounded. P_n increasing sequence in

$\mathbb{Q} \cap [0,1]$ s.t

$$P_n \to 1/\sqrt{2} \in \mathbb{R}$$

Consider

$$\mathcal{G} = \left\{(-1, P_1) \cap \mathbb{Q}, (-1, P_2) \cap \mathbb{Q}, \cdots, (-1, P_n) \cap \mathbb{Q}, \cdots, (1/\sqrt{2}, 10) \cap \mathbb{Q}\right\}$$

Proposition 1.21. *$K \subseteq X$ is compact K is sequentially compact.*

Proof.

- Let $E \subseteq K$ infinite, suppose that $E' \cap K = \varnothing$. Take $x \in K \implies x \notin E'$, therefore $\exists \delta_x > 0$ s.t

$$N^*_{\delta_x}(x) \cap E = \varnothing$$

Let $\mathcal{G} = \{N_{\delta_x}(x)\}_{x \in K}$, is an open cover of K compact, hence it has a finite subcover

$$K \subseteq N_{\delta_{x_1}}(x_1) \cup \cdots \cup N_{\delta_{x_1}}(x_k)$$

But $E \subset K \implies E = (E \cap K)$ hence

$$\begin{aligned} E &\subseteq (N_{\delta_{x_1}}(x_1) \cap E) \cup \cdots \cup (N_{\delta_{x_1}}(x_k) \cap E) \\ &\subseteq \{x_1, \cdots, x_k\} \end{aligned}$$

therefore E is finite, contradiction.

□

1.2 Complete sets

Definition 1.22. Given a sequence x_n, we say that x_n is Cauchy iff for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ s.t

$$d(x_n, x_m) < \varepsilon \quad \text{for } n, m \geq N$$

Proposition 1.23. *If x_n converges, then x_n is Cauchy.*

Proof. Triangle inequality. □

Remark 1.24. The converse is false. In $(\mathbb{Q}, |.|)$, in $(\mathbb{R}, |.|)$ there exists some sequence $x_n \in \mathbb{Q}$ s.t

$$x_n \to \sqrt{2} \quad \text{as } n \to \infty$$

But $x_n \to \sqrt{2}$ implies that x_n is Cauchy in $(\mathbb{R}, |.|)$, therefore x_n is Cauchy in $\mathbb{Q}$. Suppose x_n converges to P in $(\mathbb{Q}, |.|)$, therefore $x_n \to p \in (\mathbb{R}, |.|)$ by uniqueness of limit, $p = \sqrt{2}$, contradiction.

Definition 1.25. (X, d) metric space and $K \subseteq X$, we say that K is complete iff every Cauchy Sequence in K converges in K.

Example 1.26. $(\mathbb{R}, |.|)$ is complete (Math 210)

Proposition 1.27. *If K is complete then K is closed.*

Proof. Let x_n be a sequence in K s.t $x_n \to x_0 \in X$. Since x_n converges, then x_n is Cauchy and K is complete then x_n converges in K, therefore $x_0 \in K$. □

Proposition 1.28. *K complete and $E \subseteq K$ is closed, then E is complete.*

Proof. Let x_n be a Cauchy sequence in E, then x_n is Cauchy in K which is complete, therefore $x_n \to x_0 \in K$. But since E is closed, we get that $x_0 \in E$. □

Proposition 1.29. *(X, d) metric space and x_n Cauchy sequence in $K \subseteq X$. If x_n has a convergent subsequence x_{n_k} then x_n converges to the subsequential limit.*

Proof. Triangle inequality. □

Proposition 1.30. *(X, d) metric space and $K \subseteq X$ is sequentially compact then K is complete (and therefore closed).*

Proof. Let $\{x_n\}$ be a Cauchy sequence in K, K is sequentially compact therefore x_n has a convergent subsequence x_{n_k} that converges in K. Therefore by the previous proposition we get that x_n converges in K. □

Theorem 1.31. *Let V be a finite dimensional inner product space over $\mathbb{R}$, then V is complete w.r.t norm induced by the inner product. Therefore $K \subseteq V$ is complete $\iff$ K is closed.*

(For the converse $K \subseteq V$ which is closed, then K is complete).

Proof. Let $S = \{v_1, \cdots, v_l\}$ be an orthonormal basis of V, take x_n be a cauchy sequence in V. Therefore

$$x_n = r_{1,l} v_1 + \cdots + r_{l,n} v_l$$

Take $r_{1,n} = \langle x_n, v_1 \rangle$ therefore

$$\begin{aligned} |r_{1,m} - r_{1,n}| &= |\langle x_m, v_1 \rangle - \langle x_n, v_1 \rangle| \\ &= |\langle x_m - x_n, v_1 \rangle| \\ &\leq \|x_m - x_n\| \|v_1\| \quad \text{By C.S} \end{aligned}$$

Let $\varepsilon > 0$ then $\exists N \in \mathbb{N}$ s.t

$$\|x_m - x_n\| < \varepsilon \implies r_{1,n} \text{ is Cauchy in } \mathbb{R}$$

THerefore $r_{1,n}$ converges to $r_1 \in \mathbb{R}$. Do the same fore very coordinate, and define

$$x = r_1 v_1 + \cdots + r_n v_n$$

Therefore

$$\begin{aligned} \|x_n - x\| &= \|(r_{1,n} - r_1) v_1 + \cdots + (r_{1,l} - r_l) v_l\| \\ &\leq \|r_{1,n} - r_1\| \|v_1\| + \cdots + \|r_{l,n} - r_l\| \|v_l\| \end{aligned}$$

Therefore

$$x_n \to x$$

□

1.3 Total Boundedness

Definition 1.32. (X, d) metric space and $K \subseteq X$ we say that K is totally bounded iff for every $\varepsilon > 0$ $\exists x_1, \cdots, x_t \in K$ s.t

$$K \subseteq N_\varepsilon(x_1) \cup \cdots \cup N_\varepsilon(x_t)$$

Definition 1.33. $K \subseteq X$ is bounded iff $\exists R > 0$ s.t

$$K \subseteq N_R(x_0) \quad \text{for some } x_0 \in K$$

We say that a sequence is bounded iff

$$E = \{x_n\}_{n \in \mathbb{N}} \quad \text{is bounded}$$

Proposition 1.34. *Cauchy sequences are bounded.*

Proposition 1.35. *$K \subseteq X$ is totally bounded it is bounded.*

Note that the converse is false.

Example 1.36. $(\mathbb{R}, d_{disc})$, $\mathbb{R}$ is bounded since

$$\mathbb{R} \subseteq N_2(0)$$

but it isn't totally bounded, take $\varepsilon = 1/2$ then if it is totally bounded, then $\exists x_1 \cdots x_t \in \mathbb{R}$ s.t

$$\mathbb{R} \subseteq N_{1/2}(x_1) \cup \cdots \cup N_{1/2}(x_t) = \{x_1, \cdots, x_t\}$$

Proof. Take $\varepsilon = 1$ then $K \subseteq N_1(x_1) \cup N_1(x_t)$ for $x_1, \cdots, x_t \in K$. Take x_1 and let $d = \max_{k \in \{1, \cdots, t\}}(x_1, x_t)$, therefore for any $x \in K$ we have that

$$d(x, x_1) \leq d(x, x_i) + d(x_i, x_1)$$

for some x_i s.t $x \in N_1(x_i)$ (Exists by the above), therefore

$$d(x, x_1) \leq 1 + d \implies x \in N_{1+d}(x_1)$$

□

Proposition 1.37. *K is sequentially compact then K is totally bounded.*

Proof. Suppose not, then $\exists \varepsilon_0$ s.t K cannot be covered by finitely many ε_0-neighborhoods. Fix some x_1 then $\exists x_2 \in K \setminus N_{\varepsilon_0}(x_1)$, Similarly there exists some $x_3 \in K \setminus N_{\varepsilon_0}(x_1) \cup N_{\varepsilon_0}(x_2)$ and in general

$$\exists x_n \in K \setminus (N_{\varepsilon_0}(x_1) \cup \cdots \cup N_{\varepsilon_0}(x_{n-1}))$$

Therefore we have that for $n \neq m$

$$d(x_n, x_m) \geq \varepsilon_0$$

Therefore x_n doesn't have a convergent subsequence; contradiction. □

Lemma 1.38. *Every subset of a totally bounded set is totally bounded.*

Proof. $K \subseteq X$ totally bounded, and $E \subseteq K$; take $\varepsilon > 0$, therefore $\exists x_1, x_2, \cdots, x_t \in K$ s.t

$$K \subseteq N_{\varepsilon/2}(x_1) \cup \cdots \cup N_{\varepsilon/2}(x_t)$$

Ignore the neighborhood that don't intersect E, renumbering we have $x_1, \cdots, x_l$ s.t

$$N_{\varepsilon/2}(x_i) \cap E \neq \varnothing$$

We get

$$E \subseteq N_{\varepsilon/2}(x_1) \cup \cdots \cup N_{\varepsilon/2}(x_l)$$

Take $y_i \in N_{\varepsilon/2}(x_i) \cap E$, then

$$N_\varepsilon(y_i) \supseteq N_{\varepsilon/2}(x_i)$$

Therefore $E \subseteq N_\varepsilon(y_1) \cup \cdots \cup N_\varepsilon(y_l)$ and hence E is totally bounded. □

Example 1.39. $[0, 1]$ is totally bounded (since it is compact), therefore $(0, 1)$ is totally bounded.

Proposition 1.40. (X, d) *Metric space,* $K \subseteq X$ *is complete and totally bounded then* K *is compact.*

Proof. Assume K is not compact, fix $\mathcal{G} = \{G_\alpha\}_{\alpha \in I}$ an open cover of K with no finite subcover. Let $\varepsilon = 1$, K is totally bounded then K can be covered by finitely many 1-balls with center in K. Hence $\exists x_1 \in K$ s.t $K_1 = N_1(x_1) \cap K$ cannot be covered by finitely many open sets in $\mathcal{G}$. But K_1 is totally bounded (subset of K), now let $\varepsilon = 1/2$, $\exists x_2 \in K_1$ s.t $K_2 = N_{1/2}(x_2) \cap K_1$ cannot be covered by finitely many open sets in $\mathcal{G}$. Similarly, we construct a sequence x_n and

$$K_n = N_{1/n}(x_n) \cap K_{n-1} \quad x_n \in K_{n-1}$$

such that K_n cannot be covered by finitely many G_α's. Notice that

$$K_0 = K \supseteq K_1 \supseteq K_2 \supseteq \cdots$$

Claim. x_n is Cauchy. Indeed, let $\varepsilon > 0$ then $\exists N$ s.t

$$1/N < \varepsilon/2$$

Now take $m, n > N$, then

$$x_n \in K_{n-1} \subseteq K_N, \quad x_m \in K_{m-1} \subseteq K_N$$

therefore $x_n, x_m \in K_N = N_{1/N}(x_N) \cap K_{n-1}$, and hence

$$d(x_n, x_m) \leq 2/N < \varepsilon$$

By completeness of K, we get that x_n converges to some $x_0 \in K$; since $\mathcal{G} = \{G_\alpha\}$ is an open cover of K, there exists $\alpha_0 \in I$ s.t $x_0 \in G_{\alpha_0}$ (open set). So $\exists \varepsilon_0$ s.t

$$N_{\varepsilon_0}(x_0) \subseteq G_{\alpha_0}$$

But $x_n \to x_0$ then $\exists N_1$ s.t

$$x_N \in N_{\varepsilon_0/2}(x_0) \text{ and } 1/N < \varepsilon_0/2$$

Therefore

$$K_N \subseteq N_{1/N}(x_n) \subseteq N_{\varepsilon_0}(x_0) \subseteq G_{\alpha_0}$$

contradiction. □

Theorem 1.41. *V finite dimensional product normed space (over $\mathbb{R}$). Let $S = \{v_1, \cdots, v_n\}$ a basis of V. Then the cube*

$$Q_S = \{r_1 v_1 + r_2 v_2 + \cdots + r_n v_n \colon 0 \leq r_i \leq 1\}$$

is compact.

Proof. Let $x^k = r_{1,k} v_1 + r_{2,k} v_2 + \cdots + r_{n,k} v_n$ a sequence in Q_S, now clearly

$$r_{1,k} \text{ is a sequence in } [0,1], \text{ compact set}$$

Therefore $r_{1,k}$ has a convergent subsequence r_{1,k_n}. Similarly $r_{2,k_l} \in [0,1]$ therefore r_{2,k_l} has a convergent subsequence $r_{2,k_{l_t}}$. Doing this n times we get that $\exists$ a subsequence $\{k_j\}$ s.t

$$r_{1,k_j},\ r_{2,k_j}, \cdots,\ r_{n,k_j} \to r_1, r_2, \cdots, r_n$$

take

$$x_0 = r_1 v_1 + \cdots + r_n v_n$$

We can show that

$$\|x_0 - x^{k_j}\| \leq |r_{1,k_j} - r_1| \|v_1\| + \cdots + |r_{n,k_j} - r_n| \|v_n\| \to 0$$

then

$$x^{k_j} \to 0$$

Hence Q_S is (sequentially) compact. □

Theorem 1.42. *Let V be a finite inner product space then*

$$K \subseteq V \text{ is compact} \iff K \text{ closed and bounded}$$

Proof.
- K compact $\implies$ complete $\implies$ closed and $\implies$ totally bounded $\implies$ bounded.
- K is bounded, therefore $\exists R > 0$ s.t $K \subseteq B_{\|\cdot\|}(0, R)$. Let $S = \{v_1, \cdots, v_n\}$ be a basis of V and Take $x \in K$ therefore

$$x = r_1 v_1 + \cdots + r_n v_n$$

then $r_i = \|\langle x, v_i\rangle\| \leq \|x\| \cdot \|v_i\|$

$$B_{\|.\|}(0, R) \subseteq \{r_1 v_1 + \cdots + r_n v_n \colon r_i \leq R\}$$

Therefore $K \subseteq$ some cube and hence K is compact.

□

Proposition 1.43. *In a finite dimensional inner product space we get that bounded $\iff$ totally bounded.*

Proof. $\implies$ Bounded $\implies$ can be put in a cube (compact) $\implies$ Totally bounded.
$\impliedby$ True in general.

□

1.4 Continuity

Definition 1.44. X, Y metric spaces and $E \subseteq X$ and $f \colon E \to Y$ map; $x_0 \in E'$ we say that

$$\lim_{x \to x_0} f(x) = y \in Y$$

iff for every $\varepsilon > 0$ $\exists \delta > 0$ s.t

$$f(x) \in N_\varepsilon(y_0) \text{ for } x \in N^*_\delta(x_0) \cap E$$

Definition 1.45. $f \colon E \to Y$ continuous at x_0 iff for every $\varepsilon > 0$, $\exists \delta > 0$ s.t $f(x) \in N_\varepsilon(x_0)$ for $x \in N_\delta(x_0) \cap E$; i.e

$$d_Y(f(x), f(x_0)) < \varepsilon \quad \text{for } 0 < d_X(x, x_0) < \delta$$

Remark 1.46. f is continuous at $x_0 \iff x_0$ is an isolated point or $x_0 \in E'$ and $\lim x \to x_0 f(x) = f(x_0)$.

Proposition 1.47. $\lim_{x \to x_0} f(x) = y_0$ *iff for every sequence* $x_n \to x_0$ *we have that* $f(x_n) \to y_0$. *Therefore* f *is continuous at* x_0 *iff for every* $x_n \to x_0$ *we have that* $f(x_n) \to f(x_0)$.

Corollary 1.48. *if* f, g *are continuous then* $f + g,\ cf,\ f \cdot g,\ f/g$ *are all continuous at* x_0.

Example 1.49. $f(x, y) = \frac{xy}{x^2 + y^2}$ we want to find

$$\lim_{(x,y) \to (0,0)} f(x, y)$$

Take the sequence $(x_n, y_n) = (1/n, 0)$ then

$$\lim_{n \to \infty} (x_n, y_n) = (0, 0)$$

and

$$f(x_n, y_n) = 0 \to 0 \text{ as } n \to \infty$$

Now take $(x_n, y_n) = (1/n, 1/n)$, then

$$\lim_{n\to\infty}(x_n, y_n) = (0,0)$$

but

$$f(x_n, x_n) = \frac{1/n \cdot 1/n}{1/n^2 + 1/n^2} = 1/2 \to 1/2 \text{ as } n \to \infty$$

Therefore

$$\lim_{n\to\infty} f(x,y) \text{ does not exist}$$

Now we do the same for

$$f(x,y) = \frac{x^2 y}{x^2 + y^2}$$

Notice that

$$0 \leq |f(x,y)| = |\frac{x^2}{x^2+y^2}y| \leq |y|$$

but

$$\lim_{(x,y)\to(0,0)} |y| = 0$$

therefore by squeeze we get that

$$\lim_{(x,y)\to 0} |f(x,y)| = 0$$

Hence

$$\lim_{(x,y)\to 0} f(x,y) = 0$$

Some useful inequalities

$$x^2 + y^2 \geq x^2$$
$$x^2 + y^2 \geq y^2$$
$$x^2 + y^2 \geq 2(xy)$$

Example 1.50. Let $(V, \|.\|)$ a normed space and

$$\varphi\colon V \to \mathbb{R}, \quad \varphi(x) = \|x\|$$

Then φ is continuous.

Let x_n a sequence in V that converges to x_0 in $(V, \|.\|)$ therefore

$$\|x_n - x_0\| \to 0$$

We want to show that $\varphi(x_n) \to \varphi(x_0)$ i.e that

$$|\|x_n\| - \|x_0\|| = 0$$

But as shown in PS1, we have that

$$|\|x_n\| - \|x_0\|| \leq \|x_n - x_0\|$$

Proposition 1.51. *$f\colon X \to Y$ is continuos iff $f^{-1}(G)$ open in X for every G open in Y iff $f^{-1}(H)$ is closed in X for every H closed in Y.*

Proof. $\implies$ let x_n be a sequence in $f^{-1}(H)$ that converges to x_0 in X. Hence $f(x_n)$ is a sequence in H by continuity of f we have that

$$\lim_{n\to\infty} f(x_n) = f(x_0)$$

Moreover since H is closed, we get that $f(x_0) \in H$. Hence

$$x_0 \in f^{-1}(H)$$

Therefore if f is continuous $f^{-1}(G)$ is open in X for every G open in Y.

$\impliedby$ Now we show that if $f^{-1}(G)$ is open in X for all G open in Y then f is continuous. Fix $x_0 \in X$, let $\varepsilon > 0$ then consider the open set in Y $N_\varepsilon(f(x_0))$ hence $f^{-1}\left(N_\varepsilon(f(x_0))\right)$ is open in X. But $x_0 \in f^{-1}\left(N_\varepsilon(f(x_0))\right)$ hence $\exists \delta > 0$ s.t

$$f\left(N_\delta(x_0)\right) \subseteq N_\varepsilon(f(x_0))$$

□

Example 1.52. $E = \{x, y \in \mathbb{R}^2 \colon y \geq x\}$ is open in $\mathbb{R}^2$, define the continuous function

$$f\colon \mathbb{R}^2 \to \mathbb{R}, f(x,y) = y - x$$

Then

$$E = f^{-1}\left((0,\infty)\right)$$

which is an open set.

Example 1.53. $f\colon X \to Y$ continuous and consider the c-level set with $c \in Y$

$$E = \{x \in X \colon f(x) = c\} = f^{-1}\left(\{c\}\right)$$

is closed in X.

Therefore $E = \{(x,y) \colon x^2 + 3y^2 = 1\}$ is closed in $\mathbb{R}^2$, S^n is closed.

Remark 1.54. V, W normed spaces and

$$\|.\|_a,\ \|.\|_b \text{ equivalent norms on } V$$

so $f\colon (V, \|.\|_a) \to W$ is continuous iff $f\colon (V, \|.\|_b) \to W$ is continuous.

Equivalently: V, W normed spaces and

$$\|.\|_a,\ \|.\|_b \text{ equivalent norms on } W$$

so $f\colon V \to (W, \|.\|_a)$ is continuous iff $f\colon V \to (W, \|.\|_b)$ is continuous.

Example 1.55. Take $f\colon \mathbb{R}^m \to \mathbb{R}^n$ defined as follows

$$f(x) = (f_1(x), \cdots, f_n(x)) \quad x = (x_1, \cdots, x_m)$$

Then f is continuous iff $f_i\colon \mathbb{R}^m \to R^n$ is continuous for $1 \leq i \leq n$.

Theorem 1.56 (Extreme Value Theorem)**.** *X, Y metric spaces and $K \subseteq X$ compact and $f\colon K \to Y$ continuous. Then $f(K)$ is compact.*

Corollary 1.57. *If $Y = \mathbb{R}$, then the above implies that f attains its maximum and minimum in K.*

Proof. Let y_n be a sequence in $f(K)$, therefore $\exists x_n \in K$ s.t $f(x_n) = y_n$. Since K is compact, x_n has a convergent subsequence x_{n_k} that converges to $x_0 \in K$, using continuity we get that

$$y_{n_k} \to f(x_0) \in f(K)$$

□

Theorem 1.58. *All norms in a finite dimensional vector space are equivalent.*

Proof. V is a finite dimensional vector space, then by PS1 we can define an inner product on V. Let $\|.\|$ be the norm induced by our choice of inner product. Let $S = \{v_1, \cdots, v_n\}$ an orthonormal basis of V. Let $\|.\|_a$ be an arbitrary norm on V, we will show that $\|.\|$ and $\|.\|_a$ are equivalent. Let $x \in V$, then

$$x = r_1 v_1 + \cdots + r_n v_n, \quad |r_i| = \langle x, v_i \rangle \leq \|x\|$$

Therefore $\forall x \in V$

$$\begin{aligned}
\|x\|_a &= \|r_1 v_1 + \cdots + r_n v_n\| \\
&\leq |r_1| \|v_1\|_a + \cdots + |r_n| \|v_n\|_a \\
&\leq (\|v_1\|_a + \cdots + \|v_n\|_a) \cdot \|x\| \\
&\leq c_1 \cdot \|x\|
\end{aligned}$$

Now define the function $\varphi\colon (V, \|.\|) \to \mathbb{R}$ s.t

$$\varphi(x) = \|x\|_a$$

We claim that φ is continuous, indeed let $x_n \in V$ s.t $x_n \to x_0$ in $\|.\|$. Therefore

$$\begin{aligned}|\varphi(x_n) - \varphi(x_0)| &= |\|x_n\|_a - \|x_0\|_a| \\ &\leq \|x_n - x_0\| \\ &\leq c_1\|x_n - x_0\| \to 0\end{aligned}$$

Therefore $\lim_{n\to\infty} \varphi(x_n) = \varphi(x_0)$, so φ is continuous. Let $K = \{x \in V \colon \|x\| = 1\}$ closed and bounded in the inner product space $(V, \|.\|)$. Then it is compact in $(V, \|.\|)$ therefore $\varphi(K)$ is compact in $\mathbb{R}$ so it attains its minimum in K. Notice that $\varphi(x) = 0 \iff x = 0$, hence $\forall x \in K,\ \varphi(x) > 0$. Then

$$\min_{x\in K} \varphi(x) > 0$$

Let $x \in V$ and $x \neq 0$ then $\frac{x}{\|x\|} \in K$, therefore

$$\varphi(\frac{x}{|x|}) = \|\frac{x}{\|x\|}\|_a \geq \min \varphi(K)$$

Therefore

$$\|x\|_a \geq \min(\varphi(K))\|x\|$$

□

Corollary 1.59. *If V is finite dimensional then compact iff closed and bounded (independently of the choice of norm).*

1.5 Connected sets

Definition 1.60. (X, d) metric space and $\Omega \subseteq X$, we say that Ω is disconnected iff $\exists U, V \subseteq X$ open sets s.t

$$\Omega = (U \cup V) \cap \Omega$$

With $U \cap \Omega,\ V \cap \Omega \neq \varnothing$ and $U \cap \Omega \cap V = \varnothing$.

Otherwise Ω is connected.

Example 1.61. Let $\Omega = (1, 2] \cup (3, 4)$ disconnected, since

$$\Omega \subseteq (0, 2.5) \cup (2.5, 17)$$

$\mathbb{Q}$ is disconnected in $\mathbb{R}$ since

$$\mathbb{Q} \subseteq (-\infty, \sqrt{2}) \cup (\sqrt{2}, \infty)$$

Proposition 1.62. *X is a connected metric space iff the only sets that are both open and closed are $\varnothing, X$.*

Proof. $\implies$ Assume that U is open and closed an $U \notin \{\varnothing, X\}$ then U^c is also not open and closed and not in $\{\varnothing, X\}$ but

$$X = U \cup U^c$$

Therefore X is disconnected.

$\Longleftarrow$ Assume that X is disconnected, then $X = U \cup V$ with U, V open disjoint and nonempty. Therefore $V = U^c$ and U is open then V is closed. Hence V is a clopen set of X but $V \notin \{\varnothing, X\}$ contradiction.

□

Definition 1.63. $I \subseteq \mathbb{R}$ is an interval for $x, y \in I$ s.t $x < y$ then $z \in I$, $\forall x < z < y$.

Theorem 1.64. *$\Omega \subseteq \mathbb{R}$ is connected iff Ω is an interval.*

Proof. $\Longrightarrow$ Assume that Ω is not an interval, therefore $\exists x, y \in \Omega$ s.t $x < z < y$ and $z \notin \Omega$, therefore

$$\Omega \subseteq (-\infty, z) \cup (z, \infty)$$

Therefore Ω is disconnected.

$\Longleftarrow$ Let Ω be an interval. Suppose Ω is not connected, then $\exists U, V$ open and disjoint in Ω in $\mathbb{R}$ s.t

$$\Omega \subseteq U \cup V$$

Then $\exists x \in U \cap \Omega$ and $y \in V \cap \Omega$ and assume wlog that $x < y$; therefore $[x, y] \subseteq \Omega$ consider

$$z = \sup \{[x, y] \cup U\}$$

Note that $z \in [x, y]$ therefore $z \in \Omega$ therefore either $z \in U$ or $z \in V$. Notice that $z > x$ since $x \in U$ an open set. Also notice that $z < y$ since if $z = y \in V$ we have that $\exists \varepsilon_1$ s.t

$$(y - \varepsilon, y) = (z - \varepsilon, z) \subseteq [x, y] \cap V$$

since z is the sup we get that

$$(z - \varepsilon, z) \cap [x, y] \cap U \neq \varnothing$$

contradiction.

Therefore $z \in (x, y)$, suppose that $z \in U$ then $(z, z + \varepsilon_0) \subseteq (x, y) \cap U$ contradiction because $z + \varepsilon_0/2 > z = \sup[x, y] \cap y$. Suppose that $z \in V$ then $\exists \varepsilon_1$ s.t $(z - \varepsilon_1, z) \subseteq (x, y) \cap V$. Since z is the sup then $\exists z^*$ an element s.t

$$z - \varepsilon_1 < z^* < z \text{ and } z^* \in U$$

$z^* \in U \cap V$ contradiction.

□

Theorem 1.65 (Intermediate Value Theorem). *$f \colon X \to Y$ continuous, if X is connected then $f(X)$ is connected. In particular if $Y = \mathbb{R}$ we have the following:*

- *If $y_0 < y_1 \in f(X)$ then $\forall \alpha$ s.t $y_0 < \alpha < y_1$ $f^{-1}(\{c\}) \neq \varnothing$.*

Proof. Suppose that $f(X)$ is disconnected, then $\exists U, V \subseteq Y$ open sets that separate $f(X)$. There-

fore

$$\begin{aligned} X &= f^{-1}(f(X)) \\ &= f^{-1}(U \cap V) \cap f(X) \\ &= \left[f^{-1}(U) \cup f^{-1}(V)\right] \cap X \end{aligned}$$

But f is continuous therefore $f^{-1}(U),\ f^{-1}(V)$ are open and disjoint. Now $U \cap f(X) \neq \varnothing$ therefore $f^{-1}(U) \neq \varnothing$, same for V; then X is disconnected. □

Definition 1.66. X is path connected iff for every $a, b \in X$ there exists a continuous function

$$\alpha \colon [0,1] \to X$$

s.t $\alpha(0) = a,\ \alpha(1) = b$.

Remark 1.67. Let X be a vector space, the case where for every $a, b \in X$ the line segment $(1-t)a + tb \in X\ \forall t \in [0,1]$ we say that X is convex.

Example 1.68. $B_R(x_0)$ is always convex and hence path connected in a vector space. Indeed, take $a, b \in B_R(x_0)$ consider

$$\begin{aligned} \|(1-t)a + tb - x_0\| &< (1-t)\|a - x_0\| + t\|b - x_0\| \\ &< (1-t)R + R = R \end{aligned}$$

Proposition 1.69. *(X, d) metric space then X is path connected $\implies$ X is connected.*

Proof. Let X be disconnected then we have U, V nonempty disjoint and open

$$X = U \cup V$$

Then $\exists a \in U$ and $b \in V$, since X is path connected $\exists \alpha \colon [0,1] \to X$ s.t

$$\alpha(0) = a, \quad \alpha(1) = b$$

$[0,1]$ is connected, therefore $\alpha([0,1])$ is also connected but

$$[0,1] = \alpha^{-1}(U) \cup \alpha^{-1}(V)$$

But α is continuous, therefore $\alpha^{-1}(U) \ni 0$ and $\alpha^{-1}(V) \ni V$ which is a contradiction since $[0,1]$ is connected. □

Theorem 1.70. *$(V, \|.\|)$ vector space and $\Omega \subseteq V$ open then Ω is connected iff Ω is path connected.*

Proof. $\Longleftarrow$ True in general.

$\Longrightarrow$ We will show that A is closed by showing that A is open. Take $z \in A^c$ so there is no path joining z to x_0. But Ω is open, so $\exists \delta > 0$ s.t

$$B_\delta(z) \subseteq \Omega$$

Now every point in $B_\delta(z)$ cannot be joined to x_0; Hence

$$B_\delta(z) \subseteq A^c$$

so A^c is open.
Therefore A open, closed and nonempty in Ω a connected space, hence $A = \Omega$.

□

Example 1.71 (Topological sine). $\Omega = \{(x, \sin(1/x)) \mid x > 0\} \cup \{(0,0)\}$; let

$$A = \{(x, \sin 1/x) \colon x > 0\}$$

then A is connected (image of a continuous function), with

$$\overline{A} = A \cup (\{0\} \times [-1, 1])$$

Therefore $A \subseteq \Omega \subseteq \overline{A}$.

Suppose that Ω was connected, take $x_0 > 0$, suppose that $\exists \alpha \colon [0,1] \to \Omega$ from $(0,0)$ to $(x_0, \sin 1/x_0)$. let $t^* = \sup\{t \in [0,1] \mid x(t)\} = 0$; therefore

$$x(t^*) = 0$$

with $x(t^*) < 1$, hence

$$\alpha(t^*) = (0,0)$$

Take $\varepsilon = 1$, then by the IVT $\exists t^* < t_1 < t^* + 1$ and $t_1 \in [0,1]$ s.t

$$x(t_1) = \frac{1}{(4n_1 + 1)\pi/2}$$

Take $\varepsilon = 1/2$, then $\exists t_2 \in [0,1]$ s.t $t^* < t_2 < t^* + 1/2$ s.t

$$x(t_2) = \frac{1}{(4n_2 + 1)\pi/2}$$

We take a sequence $t_k \in [0,1]$ s.t $t^* < t_k < t^* + 1/k$

$$x(t_k) = \frac{1}{(4n_k + 1)\pi/2}$$

hence

$$\alpha(t_k) = (\frac{1}{(4n_k + 1)\pi/2}, 1)$$

With $t_k \to t^*$, therefore

$$\alpha(t^*) = (0,1)$$

contradicting the continuity of α.

2 Differentiability

Review linear transformations.

Proposition 2.1. *Let V and W normed spaces $T\colon V \to W$ linear transformation, then the following are all equivalent:*

- *T is continuous on V.*
- *T is continuous on 0_V.*
- *$\exists M > 0$ s.t*

$$\|Tx\|_W \leq M\|x\|_V$$

Proof. $\implies$ 2 Trivial.

2 $\implies$ 3 T is continuous at 0, hence for $\varepsilon = 1$ $\exists \delta > 0$ s.t

$$\|Tx - T0\|_W < 1 \quad \forall \|x - 0\|_V < \delta$$

Therefore

$$\|Tx\|_W < 1 \quad \forall \|x\|_V < \delta$$

Take any $x \neq 0$ then

$$\|\frac{x}{\|x\|_V} \cdot \delta/2\|_V = \delta/2$$

Therefore

$$\|T\left(\frac{x}{\|x\|_V} \cdot \delta/2\right)\|_W < 1$$

therefore

$$\|Tx\|_W < \frac{2}{\delta}\|x\|$$

3 $\implies$ 1

$$\|Tx - Ty\|_W = \|T(x-y)\|_W \leq M\|x-y\|_V$$

□

Example 2.2 (The derivative as a linear transformation)**.** Take $C^1([0,1])$ the space of continuously differentiable functions, with the sup norm

$$\|f\|_\infty = \sup |f(x)| \colon x \in [0,1]$$

and take $C([0,1])$ to be the space of continuous functions with the sup norm. Take

$$\frac{d}{dx} \colon C^1([0,1]) \to C([0,1])$$

Note that this is a linear transformation.

Is it continuous? If it was then $\exists M$ s.t

$$\|f\|_\infty \leq M\|f\|_\infty \quad \forall f$$

take $f_n(x) = x^n$, then

$$\|f_n\|_\infty = 1$$

but $f_n'(x) = nx^{n-1}$ with

$$\|f_n'\|_\infty = n$$

Contradiction to the Archimedean property.

Theorem 2.3. *Let V be finite dimensional, and $T\colon V \to W$ linear transformation then T is continuous.*

Proof. Let $\alpha = \{v_1, \cdots, v_n\}$ be a basis of V. Let $x \in V$, then $\exists r_1, \cdots, r_n$ s.t

$$x = r_1 v_1 + \cdots + r_n v_n$$

Since all norms are equivalent in a finite dimensional vector space, wlog work with the inner product norm and make α an orthonormal basis.

$$\begin{aligned}\|Tx\|_W &= \|r_1 T v_1 + \cdots + r_n T v_n\|_W \\ &\leq |r_1|\|Tv_1\|_W + \cdots + |r_n|\|Tv_n\|\end{aligned}$$

With

$$|r_i| = |\langle x, v_i\rangle| \leq \|x\|$$

Therefore

$$\|Tx\|_W \leq \|x\|_V(\|Tv_1\| + \cdots_{\|} TV_n\|) = M\|x\|$$

□

Motivation. In one dimension, $f\colon \mathbb{R} \to \mathbb{R}$ and $x_0 \in \mathbb{R}$

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

if it exists we say that f is differentiable at x_0 and we define

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

In other words, we can say that

$$\begin{aligned}
L = \lim_{x\to x_0} \frac{f(x)-f(x_0)}{x-x_0} &\iff \lim_{x\to x_0}\left(\frac{f(x)-f(x_0)}{x-x_0} - L\right) = 0 \\
&\iff \lim_{x\to x_0}\left(\frac{f(x)-f(x_0)-L(x-x_0)}{x-x_0}\right) = 0 \\
&\iff \exists L \in \mathbb{R} \text{ s.t } f(x) = f(x_0) + L(x-x_0) + o(|x-x_0|) \text{ with } \frac{o(t)}{|t|} \to 0 \\
&\iff \exists L \in \mathbb{R} \text{ s.t } f(x) = f(x_0) + L(x-x_0) + o(|x-x_0|) \\
&\quad \text{with } \frac{o(|x-x_0|)}{|x-x_0|} \to 0 \text{ as } x \to x_0
\end{aligned}$$

Definition 2.4. V, W normed spaces and $U \subseteq V$ open set with $x_0 \in U$ we say that f is differentiable at x_0 iff there exists a continuous linear transformation $T \in \mathcal{L}(V, W)$ s.t

$$f(x) = f(x_0) + T(x-x_0) + o(\|x-x_0\|_V)$$

or

$$f(x_0+h) = f(x_0) + T(h) + o(h)$$

with $\frac{o(\|h\|)}{\|h\|_V} \to 0$ as $h \to 0$.

Iff $\exists T \in \mathcal{L}(V, W)$ s.t

$$\lim_{h\to 0} \frac{f(x_0+h) - f(x_0) - T(h)_W}{\|h\|_V} = 0$$

Remark 2.5. If such a T exists, then it is unique.

Proof. Suppose that $\exists T_1, T_2$ continuous linear transformations in $\mathcal{L}(V, W)$ s.t

$$f(x_0+h) = f(x_0) + T_1(h) + o_1(h), \quad f(x_0+h) = f(x_0) + T_2(h) + o_2(h)$$

Therefore

$$(T_1 - T_2)(h) = (o_1 - o_2)(h) = o_3(h)$$

take v s.t $\|v\| = 1$, therefore

$$\begin{aligned}
(T_2 - T_1)(\lambda v) &= \lambda o_3(\|\lambda v\|) \\
\lambda(T_2 - T_1)(v) &= = \lambda o_3(|\lambda|) \\
(T_2 - T_1)(v) &= \frac{o_3(|\lambda|)}{\lambda}
\end{aligned}$$

Therefore as $\lambda \to 0$ we get that

$$(T_2 - T_1)(v) = 0 \quad \forall v \text{ s.t } \|v\| = 1$$

Therefore $T_2 = T_1$ □

Definition 2.6. If f is differentiable at x_0, then we let $Df(x_0)$ to be the linear transformation in

$\mathcal{L}(V, W)$ s.t

$$f(x) = f(x_0) + Df(x_0)(x - x_0) + o(\|x - x_0\|) \iff \lim_{x \to x_0} \frac{\|f(x) - f(x_0) - Df(x - x_0)\|_W}{\|x - x_0\|_V}$$

Notation. $\mathbb{R}^n = \{x = (x_1, \cdots, x_n) \colon x_i \in \mathbb{R}\}$ as rows vectors, observe $f \colon \mathbb{R}^{1\times n} \to \mathbb{R}^{m\times 1}$ differentiable at x_0 then we have that

$$f(x) = f(x_0) + Df(x_0)(x - x_0) + o(|x - x_0|)$$

Then

$$Df(x_0) \in \mathcal{L}(\mathbb{R}^m, R^n)$$

Then $Df(x_0)$ can be represented by a matrix called the *the Jacobian matrix* $[Df(x_0)]$ (an $m \times n$ matrix). Therefore

$$Df(x_0)(x - x_0) = [Df(x_0)](x - x_0)^t$$

Note that we take a row vector and output a column vector.

Example 2.7. In 1 variable we have that

$$[Df(x_0)] = [f'(x_0)]$$

Then

$$Df(x_0) \colon t \to [f'(x_0)] \cdot t = f'(x_0) \cdot t$$

Find the derivative

$$f(x, y) = x^2 + y^2$$

we get that

$$\begin{aligned} f((x, y) + (h_1, h_2)) &= f(x_0 + h_1, y_0 + h_2) \\ &= (x_0 + h_1)^2 + (y_0 + h_2)^2 \\ &= x_0^2 + y_0^2 + 2y_0h_1 + 2y_0h2 + h_1^2 + h_2^2 \\ &= f(x_0, y_0) + (2x_0, 2y_0)(h_1, h_2) + o(|h|) \end{aligned}$$

since $\frac{h_1^2+h_2^2}{\sqrt{h_1^2+h_2^2}} \to 0$ so we get

$$Df(x_0, y_0) \colon (v_1, v_2) \to (2x_0, 2y_0) \cdot (v_1, v_2)$$

therefore we get

$$[Df(x_0)] = \begin{bmatrix} 2x_0 & 2y_0 \end{bmatrix}$$

Example 2.8. Take $T \colon V \to W$ be a continuous linear transformation we want to find $DT(x)$; but

$$T(x_0 + h) = T(x_0) + T(h) + 0$$

therefore

$$DT(x_0) = T$$

Proposition 2.9. *$f\colon V \to W$ if f is differentiable at x_0 therefore f is continuous at x_0.*

Proof. Let $f(x) = f(x_0) + Df(x_0)(h) + o(\|x - x_0\|)$ but notice that

$$o(\|x - x_0\|) \to 0$$

but $Df(x_0)$ is continuous hence $\exists M$ s.t

$$\|Df(x_0)(x - x_0)\| \leq M\|x - x_0\| \to 0 \implies \lim_{x \to x_0} f(x) = f(x_0)$$

therefore f is continuous. □

2.1 Partial derivatives

Definition 2.10. $U \subseteq \mathbb{R}^n$ open and $f\colon U \to \mathbb{R}$, then

$$\frac{\partial f}{\partial x_i}(x) = \lim_{\substack{h \to 0 \\ h \in \mathbb{R}}} \frac{f(x + he_i) - f(x)}{h}$$

We call this the partial derivative w.r.t x_i.

Example 2.11.

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

We have shown that this is not continuous at $(0, 0)$ hence f is not differentiable at $(0, 0)$. But notice that

$$\frac{f(h, 0) - f(0, 0)}{h} = 0 \to 0 \text{ as } h \to 0$$

therefore $\frac{\partial f}{\partial x}(0, 0)$ exists and equal to 0. Similarly we get that $\frac{\partial f}{\partial y}(0, 0)$ exists and equal to 0.

What if $f\colon U \to \mathbb{R}$ is differentiable at x_0 then $Df(x_0) \in \mathcal{L}(R^n, \mathbb{R})$ hence $[Df(x_0)]$ is a $1 \times n$ matrix.

$$f(x_0 + he_1) = f(x_0) + Df(x_0)(he_1) + o(|h|)$$

therefore

$$\frac{f(x_0 + he_1) - f(x_0)}{h} = Df(x_0)e_1 + \frac{o(h)}{h} \to Df(x_0)e_1$$

therefore $\frac{\partial f}{\partial x_1}$ exists and is the first entry in $[Df(x_0)]$. Similarly we get $\frac{\partial f}{\partial x_i}$ exists and is the i'th entry in $[Df(x_0)]$. We get that

$$[Df(x_0)] = (\frac{\partial f}{\partial x_1}, \cdots, \frac{\partial f}{\partial x_n})$$

this is called the gradient of f denoted ∇f.

If f is differentiable then

$$\begin{aligned} f(x) &= f(x_0) + \nabla f(x_0)(x_0 - x)^t + o(|x - x_0|) \\ &= f(x_0) + \nabla f(x_0) \cdot (x - x_0) + o(|x - x_0|) \end{aligned}$$

Theorem 2.12. *$U \subseteq \mathbb{R}^n$ open, and $f : U \to \mathbb{R}$ is differentiable iff all partial derivatives exist and*

$$\lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - \nabla(x_0) \cdot h\|}{|h|} = 0$$

Proof. $\implies$ Suppose that f is differentiable, then

$$f(x_0 + h) = f(x_0) + Df(x_0)h + o(h)$$

Therefore

$$f(x_0 + he_i) = f(x_0) + Df(x_0)he_i + o(h)$$

Hence

$$\lim_{h \to 0} \frac{f(x_0 + he_i - f(x_0))}{\|h\|} = Df(x_0)e_i$$

Therefore all partial derivatives exist with

$$\nabla f(x_0) \cdot e_i = Df(x_0)e_i$$

Hence

$$\lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - \nabla(x_0) \cdot h\|}{|h|} = 0$$

$\impliedby$ Suppose that all partial derivatives exist and

$$\lim_{h \to 0} \frac{\|f(x_0 + h) - f(x_0) - \nabla(x_0) \cdot h\|}{|h|} = 0$$

We claim that

$$[Df(x_0)] = \nabla(x_0)$$

Indeed, notice that by definition

$$f(x_0 + h) - f(x_0) - \nabla(x_0) \cdot h = o(h)$$

Hence

$$f(x_0 + h) = f(x_0) + \nabla(x_0) \cdot h + o(h)$$

□

Example 2.13. $f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{o.w} \end{cases}$ then

$$f_x(0, 0) = 0, \quad f_y(0, 0) = 0$$

and

$$\frac{f(h_1,h_2)-f(0,0)-\nabla f(0,0)\cdot(h_1,h_2)}{\sqrt{h_1^2+h_2^2}}=\frac{h_1^2h_2}{(h_1^2+h_2^2)\sqrt{h_1^2+h_2^2}}$$

take $h_1=1/n, h_2=1/n$ we get that

$$\frac{1}{1\sqrt{2}}\not\to 0$$

hence this is not differentiable.

Example 2.14. $f(x,y)=x\cdot\sqrt{y}$ for $x\in\mathbb{R},\ y\geq 0$ then

$$\frac{\partial f}{\partial x}=\sqrt{y}\quad x\in\mathbb{R},\ y\geq 0$$

$$\frac{\partial f}{\partial y}=\frac{x}{2\sqrt{y}}\quad x\in\mathbb{R}, y>0$$

But what happens to $\frac{\partial f}{\partial y}$ at $(x,0)$?

$$\begin{aligned}\frac{\partial f}{\partial y}(x,0)&=\lim_{h\to 0}\frac{f(x,h)-f(x,0)}{h}\\&=\lim_{h\to 0}\frac{x\sqrt{h}}{h}\\&=\lim_{h\to 0}\frac{x}{\sqrt{h}}\end{aligned}$$

If $x\neq 0$ we have that

$$\lim_{h\to 0}\frac{x}{\sqrt{h}}\text{ doesn't exist}$$

hence $\frac{\partial f}{\partial y}$ doesn't exist at $(x,0)$ for $x\neq 0$. But for $x=0$ we have that

$$\frac{\partial f}{\partial y}(0,0)=0$$

Now we ask, is $f(x,y)=x\sqrt{y}$ differentiable at $(0,0)$. Indeed

$$\nabla f(0,0)=(f_x(0,0),f_y(0,0))=(0,0)$$

and

$$\begin{aligned}\frac{|f((0,0)+(h_1,h_2))-f(0,0)-(0,0)\cdot(h_1,h_2)|}{\|h\|}&=\frac{|h_1\sqrt{h_2}|}{\sqrt{h_1^2+h_2^2}}\\&=\sqrt{\frac{h_1^2|h_2|}{h_1^2+h_2^2}}\\&\leq\sqrt{|h_2|}\to 0\end{aligned}$$

therefore f is differentiable at $(0,0)$.

Theorem 2.15. *$U\subseteq\mathbb{R}^n$ and $f\colon U\to\mathbb{R}^n$ with $f=(f_1,\cdots,f_m)$ then f is differentiable iff*

$f_1, \cdots, f_n \colon U \to \mathbb{R}$ are all differentiable. Then

$$[Df(x_0)]_{m\times n} = \begin{pmatrix} \nabla f_1(x_0) \\ \nabla f_2(x_0) \\ \vdots \\ \nabla f_m(x_0) \end{pmatrix}$$

Therefore

$$Df(x_0)\begin{pmatrix} h_1 \\ h_2 \\ \vdots \\ h_n \end{pmatrix} = \begin{pmatrix} \nabla f_1(x_0)\cdot h \\ \nabla f_2(x_0)\cdot h \\ \vdots \\ \nabla f_m(x_0)\cdot h \end{pmatrix}$$

Example 2.16. $f(x,y) = (x^2y^2, x^2+y^2, x^2-y^2)$ then

$$[Df(x,y)] = \begin{pmatrix} 2xy^2 & 2x^2y \\ 2x & 2y \\ 2x & -2y \end{pmatrix}$$

Therefore

$$[Df(1,2)] = \begin{pmatrix} 2 & 4 \\ 2 & 4 \\ 2 & -4 \end{pmatrix}$$

Hence

$$Df(1,2)\colon \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \to \begin{pmatrix} 8h_1 + 4h_2 \\ 2h_1 + 4h_2 \\ 2h_1 - 4h_2 \end{pmatrix}$$

Proof.

- Suppose that f is differentiable, then

  $$f(x_0+h) = f(x_0) + Df(x_0)h + o(\|h\|)$$

  Taking the dot with $e_i \in \mathbb{R}^m$ we get

  $$f_i(x_0+h) = f_i(x_0) + Df(x_0)(h)\cdot e_i + o(\|h\|)$$

  Therefore f_i is differentiable with

  $$Df_i(x_0)\colon h \to (Df(x_0)(h))\cdot e_i$$

- Suppose that each f_i is differentiable then

  $$f_i(x_0+h) = f_i(x_0) + Df_i(x_0)h + o_i(\|h\|)$$

  Therefore

  $$f(x_0+h) = f(x_0) + (Df_1(x_0)h, \cdots, Df_m(x_0)h) + (o_1(\|h\|), \cdots, o_m(\|h\|))$$

Hence f is differentiable with

$$Df(x_0) = \begin{pmatrix} Df_1(x_0) \\ Df_2(x_0) \\ \vdots \\ Df_m(x_0) \end{pmatrix}$$

but $Df_i(x_0) = \nabla f_i(x_0)$. i.e

$$Df(x_0)\colon h \to (\nabla f_i(x_0) \cdot h)_{1 \leq i \leq m}$$

□

Remark 2.17. f, g differentiable, then $f + \alpha g$ is differentiable, and

$$D(f + \alpha g) = Df + \alpha Dg$$

2.2 Chain Rule.

Theorem 2.18. *Let $U \subseteq \mathbb{R}^n$ and $f\colon U \to \mathbb{R}^m$ and $V \subset \mathbb{R}^m$ open and $g\colon V \to \mathbb{R}$, take $x_0 \in U$ s.t $f(x_0) \in V$.*

If f is differentiable at x_0, and g is differentiable at $f(x_0)$. Then $h := g \circ f$ is differentiable at x_0 with

$$Dh(x_0) = D_g(f(x_0)) \circ D_f(x_0)$$

In matrix form

$$[Dh(x_0)] = [D_g(f(x_0))][Df(x_0)]$$

Example 2.19. $h\colon \mathbb{R}^m \to \mathbb{R}$ with

$$h(u, v) = g(x(u, v), y(u, v))$$

Let $f(u, v) = (x(u, v), y(u, v))$ Therefore

$$[Dh(u, v)] = [Dg(x(u, v), y(u, v))][Df(u, v)]$$
$$\begin{pmatrix} \frac{\partial h}{\partial u} & \frac{\partial h}{\partial v} \end{pmatrix} = \begin{pmatrix} \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

Therefore we get that

$$\begin{pmatrix} \frac{\partial h}{\partial u} & \frac{\partial h}{\partial v} \end{pmatrix} = \begin{pmatrix} \frac{\partial g}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial g}{\partial y}\frac{\partial y}{\partial u} & \frac{\partial g}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial g}{\partial y}\frac{\partial y}{\partial v} \end{pmatrix}$$

Example 2.20. $\gamma\colon [a, b] \to \mathbb{R}^n$ and $f\colon \mathbb{R}^n \to \mathbb{R}$ with

$$h(t) = f(\gamma(t))$$

Then

$$\begin{aligned}
[Dh(t)] &= [Df(\gamma(t))][D\gamma(t)] \\
[h'(t)] &= (\nabla f(\gamma(t)))\begin{pmatrix}\gamma_1' \\ \gamma_2' \\ \vdots \\ \gamma_n'\end{pmatrix} \\
&= \nabla f(\gamma(t)) \cdot \gamma'(t)
\end{aligned}$$

i.e $h\colon [a,b] \to \mathbb{R}$.

Consequences.

1. Mean value theorem for $\mathbb{R}^n \to \mathbb{R}$
2. $Df(x) = 0 \implies f$ is constant (in an open and connected domain).
3. If $\frac{\partial f_i}{\partial x_i}$ exist and are continuous at x_0 then f is differentiable at x_0.

Proof. f is differentiable at x_0 then

$$f(x) = f(x_0-) + Df(x_0)(x - x_0)_o(|x - x_0|) \quad (*)$$

and g is diff at $f(x_0)$, therefore

$$g(y) = g(f(x_0)) + Dg(f(x_0))(y - f(x_0)) + o(|y - f(x_0)|) \quad (**)$$

We have

$$\begin{aligned}
h(x) &= g(f(x)) \\
&= g(f(x_0)) + Dg(f(x_0))(f(x) - f(x_0)) + o(|f(x) - f(x_0)|) \\
&= h(x_0) + Dg(f(x_0))\left(Df(x_0)(x - x_0) + o(|x - x_0|)\right) + o(|f(x) - f(x_0)|) \\
&= h(x_0) + (Df(f(x_0)) \circ Df(x_0))(x - x_0) + (Dg(f(x_0))(o(|x - x_0|)) + o(|f(x) - f(x_0)|))
\end{aligned}$$

We will be done if we show that

$$R(x) = (Dg(f(x_0))(o(|x - x_0|)) + o(|f(x) - f(x_0)|)) = o(|x - x_0|)$$

but since $Dg(f(x_0))$ is a linear transformation

$$\frac{|Dg(f(x_0))(o(x - x_0))|}{|x - x_0|} = Dg(f(x_0))\left(\frac{o(x - x_0)}{|x - x_0}\right)$$

But using continuity we get as $x \to x_0$

$$Dg(f(x_0))\left(\frac{o(x - x_0)}{|x - x_0}\right) \to Dg(0) = 0$$

On the other hand, $Df(x_0)$ is a continuous linear transformation, then $\exists M$ s.t

$$\|Df(x_0)v\| \leq Mv$$

Let $\varepsilon > 0$, $\exists \delta > 0$ s.t

$$|o(|y - f(x_0)|)| < \frac{\varepsilon}{M+2}|y - f(x_0)|, \quad \forall |u - f(x_0)| < \delta$$

Now f is differentiable at x_0, then it is continuous. Hence $\exists \gamma > 0$ s.t for $|x - x_0| < \gamma$

$$|f(x) - f(x_0)| < \delta$$

Therefore for $|x - x_0| < \gamma$ we have

$$o(|f(x) - f(x_0)|) < \frac{\varepsilon}{M+2}|f(x) - f(x_0)|$$

By $(*)$ this is equal to $\frac{\varepsilon}{M+2}|Df(x_0)(x - x_0) + o(|x - x_0|)|$ therefore

$$o(|f(x) - f(x_0)|) < \frac{\varepsilon}{M+2}M\left(|x - x_0| + o(|x - x_0|)\right)$$

by choosing γ even smaller

$$o(|x - x_0|) < |x - x_0|$$

Then for $|x - x_0| < \gamma$, we get

$$\begin{aligned} o(|f(x) - f(x_0)|) &< \frac{\varepsilon}{M+2}(M+1)|x - x_0| \\ &< \varepsilon|x - x_0| \end{aligned}$$

□

Recall. $\gamma\colon [a,b] \to U \subseteq \mathbb{R}^n$ and $f\colon U \to \mathbb{R}$ differentiable. Then

$$h(t) = f(\gamma(t)) \text{ is differentiable}, \quad h\colon [a,b] \to \mathbb{R}$$

with

$$h'(t) = \nabla(\gamma(t)) \cdot \gamma'(t)$$

Theorem 2.21 (Mean Value theorem). *$U \subseteq \mathbb{R}^n$ open and convex domain, $f\colon U \to \mathbb{R}$ differentiable and $x, y \in U$ then $\exists c$ on the line segment joining x to y s.t*

$$f(y) - f(x) = \nabla f(c) \cdot (y - x)$$

Proof. Take $\gamma\colon [0,1] \to \mathbb{R}^n$ s.t

$$\gamma(t) = (1-t)x + ty$$

notice that $\gamma(1) = y \in U,\ \gamma(0) = x \in U$, therefore by convexity

$$\gamma([0,1]) \subseteq U$$

define

$$h(t) = f(\gamma(t)), \quad h\colon [0,1] \to \mathbb{R}$$

Then by MVT on $\mathbb{R}$, $\exists t_0 \in [0,1]$ s.t

$$f(y) - f(x) = f(\gamma(1)) - f(\gamma(0)) = h(1) - h(0) = h'(t_0)(1-0)$$

But $h'(t_0)$ (by chain rule)

$$\nabla f(\gamma(t)) \cdot \gamma'(t) = f(\gamma(t)) \cdot (y - x)$$

Therefore

$$f(y) - f(x) = f(\gamma(t_0)) \cdot (y - x)$$

letting $c = \gamma(t_0)$ which belongs to the line segment, as desired. □

Corollary 2.22. *Let $U \subseteq \mathbb{R}^n$ open, connected domain. Moreover, let $f\colon U \to \mathbb{R}^m$ differentiable s.t*

$$Df(x) = 0 \quad \forall x \in U$$

Then f is constant.

Proof.

- **Case $m = 1$.** U is open and connected, therefore it must be path connected. Fix $x_0 \in U$ and $y \in U$, let $A = \{x \in U\colon f(x) = f(x_0)\}$, note that $A \neq \varnothing$ and A is closed since

  $$A = f^{-1}(\{x_0\})$$

  let $x \in A$, U is open then $\exists \delta > 0$ s.t

  $$B(x,\delta) \subseteq U$$

  but $B(x,\delta)$ is convex, then by the MVT for every $y \in B(x,\delta)$ $\exists c$ in the line joining x to y s.t

  $$f(y) - f(x) = \nabla f(c) \cdot (y - x) = 0$$

  Therefore

  $$f(y) = f(x) = f(x_0)\ \forall y \in B(x,\delta)$$

  Therefore

  $$B(x,\delta) \subseteq A$$

  hence A is open. By connectedness of U we get that $A = U$.

- **General case.** Let $f = (f_1, \cdots, f_m)$ therefore

$$[Df(x)] = \begin{pmatrix} \nabla f_1 \\ \vdots \\ \nabla f_m \end{pmatrix}$$

but $\nabla f_i = 0 \implies f_i = c_i$, therefore $f = c$.

□

Theorem 2.23. *$U \subseteq \mathbb{R}^n$ and $f : U \to \mathbb{R}$ and fix $a = (a_1, \cdots, a_n) \in U$ If $\frac{\partial f}{\partial x_i}$ exist in a neighborhood of a and $\frac{\partial f}{\partial x_i}$ are continuous at a. Then f is differentiable at a.*

Corollary 2.24. *If $f : U \to \mathbb{R}^m$ and $a \in U$ if $\frac{\partial f_i}{\partial x_j}$ exist in a neighborhood of a and are continuous at a, then f is differentiable at a.*

Example 2.25. $f(x, y) = x\sqrt{y}$ is differentiable for $x \in \mathbb{R}$ and $y > 0$ and $f_x = \sqrt{y}$, $f_y = \frac{x}{2\sqrt{y}}$ which both exist and are continuous in a neighborhood of (x_0, y_0) for $x_0 \in \mathbb{R}$ and $y_0 > 0$. So f is differentiable at these points.

Proof. Let $B(a, r) \subset U$ and $|h| < R$ with $h = (h_1, \cdots, h_n)$, then

$$\begin{aligned}
f(a+h) - f(a) &= f(a + h_1e_1) - f(a) \\
&\quad + f(a + h_1e_1 + h_2e_2) + f(a + h_1e_1) \\
&\quad - f(a + h_1e_1 + h_2e_2) - f(a) \\
&\vdots \\
&\quad + f(a+h) - f(a + h_1e_1 + h_2e_2 + \cdots + h_{n-1}e_{n-1}) \\
&= \frac{\partial f}{\partial x_1}(a + c_1e_1) \cdot h_1 \\
&\quad \frac{\partial f}{\partial x_2}(a + h_1e_1 + c_2e_2) \cdot h_2 \\
&\vdots \\
&\quad + \frac{\partial f}{\partial x_n}(a + h_1e_1 + \cdots + h_{n-1}e_{n-1} + c_ne_n)h_n
\end{aligned}$$

Where $c_i \in [0, h_i]$, therefore

$$\begin{aligned}
f(a+h) - f(a) &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(a + h_1e_1 + \cdots + h_{i-1}e_{i-1} + c_ie_i)h_i \\
&= \frac{|f(a+h) - f(a) - \nabla f(a) \cdot h|}{h} \\
&= \frac{1}{|h|} \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}(a + h_1e_1 \cdots + h_{i-1}e_{i-1} + c_ie_i) - \frac{\partial f}{\partial x_i}(a) \right) h_i| \\
&\leq \sum_{i=1}^{n} \sum_{i=1}^{n} | \frac{\partial f}{\partial x_i}(a + h_1e_1 \cdots + h_{i-1}e_{i-1} + c_ie_i) - \frac{\partial f}{\partial x_i}(a)| \frac{|h_i|}{|h|} \\
&\leq \sum_{i=1}^{n} \sum_{i=1}^{n} | \frac{\partial f}{\partial x_i}(a + h_1e_1 \cdots + h_{i-1}e_{i-1} + c_ie_i) - \frac{\partial f}{\partial x_i}(a)|
\end{aligned}$$

Let $h \to 0$ then $a + h_1 e_1 + \cdots + h_{i-1} e_{i-1} + c_i e_i \to a$. and use continuity to get

$$\lim_{h \to 0} \frac{|f(a+h) - f(a) - \nabla f(a) \cdot h}{|h|} = 0$$

so f is differentiable at a. □

Remark 2.26. The converse is not true, since $f(x, y) = x\sqrt{y}$ is differentiable at $(0, 0)$ but f_y doesn't exist at $(x, 0)$ $\forall x \neq 0$.

Corollary 2.27. *$U \subset \mathbb{R}^n$ and $f \colon U \to \mathbb{R}^m$ differentiable,*

$$Df \colon U \to \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$$

Then Df is continuous on U iff all partial derivatives $\frac{\partial f_i}{\partial x_i}$ of f are continuous on U.

Proof. $Df(u) = \begin{pmatrix} \nabla f_1 \\ \nabla f_2 \\ \vdots \\ \nabla f_n \end{pmatrix}$ which is continuous iff every entry is continuous. □

Definition 2.28. V, W finite dimensional vector space $U \subset V$ open and $f \colon U \to W$ we say that f is C^1 iff f is differentiable and Df is continuous.

Note that this is equivalent to saying that all partial derivatives exist and are continuous.

2.3 Higher order derivatives.

Motivation. Let $\alpha = \{e_1, \cdots, e_n\}$ basis of $\mathbb{R}^n$ and $\hat{\alpha} = \{\hat{e_1}, \cdots, \hat{e_m}\}$ basis of $\mathbb{R}^m$. Take

$$f \colon U \to \mathbb{R}^m, \quad \text{differentiable}$$

and

$$Df \colon U \to \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$$

Define

$$\gamma = \{T_{11}, \cdots, T_{1m}, T_{21}, \cdots, T_{nm}\}$$

with

$$T_{i_0 j_0} \colon e_{i_0} \to \hat{e_{j_0}},\ e_{i \neq i_0} \to 0$$

Take $x \in U$ then

$$Df(x) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$$

with

$$Df(x) = r_{11} T_{11} + r_{12} T_{12} + \cdots + r_{nm} T_{nm}$$

Then

$$Df(x) e_j = r_{j1} \hat{e_j} + \cdots + r_{jm} \hat{e_m}$$

therefore

$$[Df(x)] = \begin{pmatrix} \nabla f_1 \\ \vdots \\ \nabla f_m \end{pmatrix}$$

Therefore

$$Df(x)e_i = \begin{pmatrix} \frac{\partial f_1}{\partial x_i} \\ \vdots \\ \frac{\partial f_m}{\partial x_i} \end{pmatrix}$$

So we get that

$$r_{ij} = \frac{\partial f_j}{\partial x_i}$$

Therefore

$$[Df]_\gamma = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} \\ \frac{\partial f_1}{\partial x_2} \\ \vdots \\ \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

Hence we get that

$$D[Df] = \begin{pmatrix} \nabla (f_1)_{x_1} \\ \nabla (f_1)_{x_2} \\ \vdots \\ \nabla (f_m)_{x_m} \end{pmatrix}$$

Note that

$$Df \colon U \to \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$$

Hence

$$D(Df) \colon U \to \mathcal{L}(\mathbb{R}^n, \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m))$$

Therefore $D[Df]$ is a $mn \times n$ matrix, which we call the Hessian matrix.

Example 2.29. $f(x, y) = (x^2y^3, x^2 + y^3, x^3 + y^2)$ we have that

We get that

$$D^2(f) = \begin{pmatrix} (f_1)_{xx} & (f_1)_{xy} \\ (f_1)_{yx} & (f_1)_{yy} \\ (f_2)_{xx} & (f_2)_{xy} \\ (f_2)_{yx} & (f_2)_{yy} \\ (f_3)_{xx} & (f_3)_{xy} \\ (f_3)_{yx} & (f_3)_{yy} \end{pmatrix}$$

Remark 2.30. In the particular case where $m = 1$ we have that $f \colon U \to \mathbb{R}$ therefore

$$D^f = \begin{pmatrix} \nabla f_{x_1} \\ \nabla f_{x_2} \\ \vdots \\ \nabla f_{x_n} \end{pmatrix}$$

In the $n = 2$ case we get

$$\begin{pmatrix} \frac{\partial}{\partial x}\frac{\partial f}{\partial x} & \frac{\partial}{\partial y}\frac{\partial f}{\partial x} \\ \frac{\partial}{\partial x}\frac{\partial f}{\partial y} & \frac{\partial}{\partial y}\frac{\partial f}{\partial y} \end{pmatrix}$$

Theorem 2.31 (Mixed derivatives theorem)**.** *$U \subseteq \mathbb{R}^n$ and $f\colon U \to \mathbb{R}$, take $x_0 \in U$. If $\frac{\partial f}{\partial x_i \partial x_j}$ and $\frac{\partial f}{\partial x_i \partial x_j}$ exist in a neighborhood of x_0 and are continuous at x_0 then they are equal.*

Proof. $n = 2$, assume that $\frac{\partial f}{\partial x_i \partial x_j}$ and $\frac{\partial f}{\partial x_i \partial x_j}$ exist in a neighborhood of a and are continuous at $a = (a_1, a_2)$. Define for $h, k > 0$

$$w(h,k) = f(a_1 + h) - f(a_1 + h, a_2) - f(a_1, a_2 + h) + f(a_1, a_2)$$

Define

$$v(h,k) = f(a_1 + h, a_2 + k) - f(a_1 + h, a_2)$$

and

$$u(h,k) = f(a_1 + h, a_2 + k) - f(a_1, a_2 + k)$$

Notice that by MVT

$$\begin{aligned} w(h,k) &= v(h,k) - v(0,k) \\ &= v_h(c,k) \quad c \in [0,h] \\ &= \left(f_{x_1}(a_1 + c, a_2 + k) - f_{x_1}(a_1 + c, a_2)\right) h \\ &= \left(f_{x_1 x_2}(a_1 + c, a_2 + d)\right) kh \quad d \in [0,k] \end{aligned}$$

Similarly

$$\begin{aligned} w(h,k) &= u(h,k) - u(h,0) \\ &= u_k(h,d')k \quad d' \in [0,k] \\ &= \left(f_{x_2}(a_1 + h, a_2 + d') - f_{x_2}(a_1, a_2 + d')\right) k \\ &= \left(f_{x_2 x_1}(a_1 + c', a_2 + d')\right) kh \quad c \in [0,h] \end{aligned}$$

Hence

$$f_{x_1 x_2}(a_1 + c, a_2 + d') = f_{x_2 x_1}(a_1 + c', a_2 + d')$$

then let $h, k \to 0$ by continuity we get

$$f_{x_1 x_2}(a_1, a_2) = f_{x_2 x_1}(a_1, a_2)$$

□

Definition 2.32. We say that $f \in C^k(U)$ ($k \in \mathbb{N}$) iff all partial derivatives of order k of f (of its components exist) and are continuous.

Taylor's theorem. In one variable, if g is C^1 then

$$g(t_0 + h) = g(t_0) + g'(t_0)h + o(|h|).$$

If g is C^2, then

$$g(t_0 + h) = g(t_0) + g'(t_0)h + \frac{g''(t_0)}{2!}h^2 + o(|h|^2).$$

More generally, if g is C^k, then

$$g(t_0 + h) = \sum_{i=0}^{k} \frac{g^{(i)}(t_0)}{i!}h^i + o(|h|^k).$$

Now take $U \subset \mathbb{R}^n$ open and a function

$$f : U \to \mathbb{R}.$$

Fix $x_0 \in U$ and assume f is C^1 in U. Take $\hat{u}$ a unit vector in $\mathbb{R}^n$, and define

$$g(t) = f(x_0 + t\hat{u}),$$

which is defined for $|t| \leq \delta$ small enough. Therefore

$$g \in C^1([-\delta, \delta]).$$

Hence

$$g(t) = g(0) + \frac{g'(0)}{1!}\, t + o(|t|).$$

But

$$g'(t) = \nabla f(x_0 + t\hat{u}) \cdot \hat{u},$$

so

$$g'(0) = \nabla f(x_0) \cdot \hat{u}.$$

Therefore

$$f(x_0 + t\hat{u}) = f(x_0) + \left(\nabla f(x_0) \cdot \hat{u}\right) t + o(|t|).$$

Now for $h \in \mathbb{R}^n$ small, write $h = |h|\hat{u}$ where $\hat{u} = \frac{h}{|h|}$ (for $h \neq 0$). Taking $t = |h|$ above gives

$$\begin{aligned} f(x_0 + h) &= f\left(x_0 + |h|\frac{h}{|h|}\right) \\ &= f(x_0) + \left(\nabla f(x_0) \cdot \tfrac{h}{|h|}\right)|h| + o(|h|) \\ &= f(x_0) + \nabla f(x_0) \cdot h + o(|h|) \\ &= f(x_0) + \sum_{i=1}^{n} f_{x_i}(x_0)\, h_i + o(|h|). \end{aligned}$$

Assume now that f is C^2. Then g is C^2 and

$$g(t) = g(0) + g'(0)t + \frac{g''(0)}{2}t^2 + o(|t|^2).$$

We have

$$g'(t) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(x_0 + t\hat{u})\,\hat{u}_i,$$

hence differentiating once more,

$$\begin{aligned} g''(t) &= \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}(x_0 + t\hat{u}) \right)' \hat{u}_i \\ &= \sum_{i=1}^{n} \left(\nabla \left(\frac{\partial f}{\partial x_i} \right)(x_0 + t\hat{u}) \cdot \hat{u} \right) \hat{u}_i \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 f}{\partial x_j \partial x_i}(x_0 + t\hat{u})\,\hat{u}_j \hat{u}_i. \end{aligned}$$

Therefore

$$g''(0) = \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 f}{\partial x_j \partial x_i}(x_0)\,\hat{u}_j \hat{u}_i.$$

Let $D^2 f(x_0)$ denote the Hessian matrix

$$D^2 f(x_0) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(x_0) \right)_{1 \le i,j \le n}.$$

Then the previous expression can be written as the quadratic form

$$g''(0) = \langle D^2 f(x_0)\hat{u}, \hat{u} \rangle.$$

Hence

$$g(t) = g(0) + (\nabla f(x_0) \cdot \hat{u})t + \frac{1}{2}\langle D^2 f(x_0)\hat{u}, \hat{u} \rangle t^2 + o(|t|^2).$$

Equivalently, for $h \in \mathbb{R}^n$ small (again write $h = |h|\hat{u}$),

$$f(x_0 + h) = f(x_0) + \nabla f(x_0) \cdot h + \frac{1}{2}\langle D^2 f(x_0)h, h \rangle + o(|h|^2).$$

Example ($n = 2$). If $f = f(x_1, x_2)$ and $h = (h_1, h_2)$, then

$$\begin{aligned} f(x_0 + h) &= f(x_0) + f_{x_1}(x_0)h_1 + f_{x_2}(x_0)h_2 \\ &\quad + \frac{1}{2} \begin{pmatrix} f_{x_1x_1}(x_0) & f_{x_1x_2}(x_0) \\ f_{x_2x_1}(x_0) & f_{x_2x_2}(x_0) \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \cdot \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} + o(|h|^2) \\ &= f(x_0) + f_{x_1}(x_0)h_1 + f_{x_2}(x_0)h_2 \\ &\quad + \frac{1}{2}\Big(f_{x_1x_1}(x_0)h_1^2 + (f_{x_1x_2}(x_0) + f_{x_2x_1}(x_0))h_1h_2 + f_{x_2x_2}(x_0)h_2^2 \Big) + o(|h|^2). \end{aligned}$$

(If $f \in C^2$, then $f_{x_1x_2}(x_0) = f_{x_2x_1}(x_0)$.)

This matches the pattern

$$\frac{f(x_0)}{0!} + \left(\frac{f_{x_1}(x_0)h_1}{1!} + \frac{f_{x_2}(x_0)h_2}{1!}\right) + \left(\frac{f_{x_1x_1}(x_0)h_1^2}{2!} + \frac{f_{x_1x_2}(x_0)h_1h_2}{1!\,1!} + \frac{f_{x_2x_2}(x_0)h_2^2}{2!}\right) + o(|h|^2).$$

Multi-index notation. Take $\alpha = (\alpha_1, \ldots, \alpha_n)$ where $\alpha_i \in \mathbb{N} \cup \{0\}$. Define the order of α by

$$|\alpha| := \alpha_1 + \cdots + \alpha_n,$$

and the factorial by

$$\alpha! := \alpha_1! \cdots \alpha_n!.$$

For $f \in C^{|\alpha|}$, define the α-partial derivative

$$D^\alpha f := \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \frac{\partial^{\alpha_2}}{\partial x_2^{\alpha_2}} \cdots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}} f.$$

For $x \in \mathbb{R}^n$, define

$$x^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}.$$

Example 2.33. Let $f(x, y, z) = x^2y^2z^3$. Then

$$\begin{aligned}
\frac{D^{(1,2,1)}f}{(1,2,1)!} &= \frac{1}{1!\,2!\,1!} \frac{\partial}{\partial x} \frac{\partial^2}{\partial y^2} \frac{\partial}{\partial z}(x^2y^2z^3) \\
&= \frac{1}{2} \frac{\partial}{\partial x} \frac{\partial^2}{\partial y^2}(3x^2y^2z^2) \\
&= \frac{1}{2} \frac{\partial}{\partial x}(6x^2z^2) \\
&= 6xz^2.
\end{aligned}$$

Theorem 2.34 (Taylor's theorem)**.** *Let $U \subset \mathbb{R}^n$ be open, $x_0 \in U$, and $f : U \to \mathbb{R}$ be C^k. Then*

$$f(x_0 + h) = \sum_{0 \le |\alpha| \le k} \frac{D^\alpha f(x_0)}{\alpha!} h^\alpha + o(|h|^k) \qquad (h \to 0),$$

equivalently,

$$f(x) = \sum_{0 \le |\alpha| \le k} \frac{D^\alpha f(x_0)}{\alpha!} (x - x_0)^\alpha + o(|x - x_0|^k) \qquad (x \to x_0).$$

2.4 Extremum

Definition 2.35. $f \colon U \to \mathbb{R}$ $U \subseteq \mathbb{R}^n$ and $x_0 \in U$. We say that f has a local minimum at x_0 iff $\exists \delta > 0$ s.t $f(x) \ge f(x_0)$ $\forall x \in B(x_0, \delta)$.

We say that f has a local maximum at x_0 iff $\exists \delta > 0$ s.t $f(x) \le f(x_0)$ $\forall x \in B(x_0, \delta)$

Theorem 2.36. *If f has a local min (or max) at x_0 and f is differentiable at x_0 then $\nabla f(x_0) = 0$.*

Proof. f has a local minimum at x_0, then $\exists \delta > 0$ s.t $f(x) > f(x_0)\ \forall x \in B(x_0, \delta)$. Let $|t| < \delta$ and $\hat{u}$ a unit vector; since f is differentiable at x_0, with $x_0 + t\hat{u} \in B(x_0, \delta)$. Hence

$$f(x_0 + t\hat{u}) = f(x_0) + (\nabla f(x_0) \cdot \hat{u})t + o(|t|) \geq f(x_0)$$

Therefore

$$(\nabla f(x_0) \cdot \hat{u})t + o(|t|) \geq 0$$

- if $t \geq 0$ we have that
$$\nabla f(x_0) \cdot \hat{u} + \frac{o(|t|)}{t} \geq 0$$
let $t \to 0$
$$\nabla f(x_0) \cdot \hat{u} \geq 0$$
- if $t \leq 0$ we have that
$$\nabla f(x_0) \cdot \hat{u} + \frac{o(|t|)}{t} \leq 0$$
Let $t \to 0$ we have
$$\nabla f(x_0) \cdot \hat{u} \leq 0$$

Therefore $\forall \hat{u},\ |\hat{u}| = 1$ we have that

$$\nabla f(x_0) \cdot \hat{u} = 0 \implies \nabla f(x_0) = 0$$

□

Example 2.37. Find local min/max of the following:

$$f(x, y) = x^2 + y^2$$

We have

$$\nabla f(x, y) = (2x, 2y) = (0, 0)$$

Hence

$$\begin{cases} 2x = 0 \\ 2y = 0 \end{cases} \implies (x, y) = 0$$

Therefore

$$f(0, 0)0 \leq x^2 + y^2 = f(x, y)\ \forall (x, y)$$

Therefore at $(0, 0)$ ia have a minimum.

$$f(x, y) = x^2 - y^2$$

Therefore

$$\nabla(x, y) = (2x, -2y) = 0 \implies x = y = 0$$

Notice that $f(0,0) = 0$ Take $\delta > 0$, $B(0,\delta)$ take

$$f(\delta/2, 0) = \delta^2/4 > 0, \quad f(0,\delta) = -\delta^2/4 < 0$$

Therefore $(0,0)$ is neither a local min nor a local max (it is a saddle point).

Second derivative test. Review the spectral theorem and diagonalization.

Suppose that A is symmetric and let $\lambda_1, \cdots, \lambda_n$ be the eigenvalues of A with corresponding orthonormal basis of eigenvectors $v_1, \cdots, v_n$, i.e

$$Av_i = \lambda_i v_i, \quad \langle v_i, v_j \rangle = \delta_{ij}$$

Let $v \in \mathbb{R}^n$, then we can write v as $r_1 v_1 + \cdots + r_n v_n$, therefore

$$Av = (\lambda_1 r_1)v_1 + \cdots + (\lambda_n r_n)v_n$$

Therefore we get that

$$\begin{aligned}\langle Av, v \rangle &= \langle r_1\lambda_1 v_1 + \cdots + r_n\lambda_n v_n, r_1 v_1 + \cdots + r_n v_n \rangle \\ &= r_1^2\lambda_1 + r_2^2\lambda_2 + \cdots + r_n^2\lambda_n\end{aligned}$$

Definition 2.38. Let A be an $n \times n$ symmetric matrix we say that A is positive definite iff

$$\langle Av, v \rangle > 0 \quad \forall v \neq 0$$

iff all eigenvalues of A are positive.

We say that it is positive semi-definite iff

$$\langle Av, v \rangle \geq 0 \quad \forall v \neq 0$$

iff all eigenvalues of A are nonnegative.

We define negative (semi) definite similarly.

Otherwise, we say that A is indefinite (at least two eigenvalues of opposite signs).

Remark 2.39. A is symmetric, then $\lambda_i \geq 0$. Wlog $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$, therefore

$$\lambda_1 |v|^2 \leq \langle Av, v \rangle \leq \lambda_n |v|^2$$

A is positive semi-definite $\iff$ $\exists c \geq 0$ s.t

$$\langle Av, v \rangle \geq c|v|^2$$

Definition 2.40. $U \subset \mathbb{R}^n$ and $f\colon U \to \mathbb{R}$ a C^1 function. We say that x_0 is a critical point of f iff

$$\nabla f(x_0) = 0$$

(Candidate of extremum)

We say that $f \in C(U)$ attains a saddle point at x_0 iff $\nabla f(x_0) = 0$ but f doesn't attain a local extremum at x_0.

Theorem 2.41 (Second derivative test)**.** *$U \subset \mathbb{R}^n$ open and $f\colon U \to \mathbb{R}$ a C^2 function, take $x_0 \in U$ s.t*

$$\nabla f(x_0) = 0$$

1. *If $D^2 f(x_0)$ is positive definite, then f attains a local minimum.*
2. *If $D^2 f(x_0)$ is negative definite, then f attains a local maximum.*
3. *If $D^2 f(x_0)$ has two eigenvalues with opposite signs, then f has a saddle point at x_0.*

The test is inconclusive if $Df(x_0)$ is semi-definite.

Proof. By Taylor's theorem, we have that

$$f(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2}\langle D^2 f(x_0)(x - x_0), x - x_0\rangle + o(|x - x_0|^2)$$

(Since $\nabla f(x_0) = 0$) we get

$$\begin{aligned}
f(x) - f(x_0) &= 1/2\langle D^2 f(x_0)(x - x_0), x - x_0\rangle + o(|x - x_0|^2) \\
&\geq C|x - x_0|^2 + o(|x - x_0|^2) && C > 0 \\
&= |x - x_0|^2 \left(C + \frac{o(|x - x_0|^2)}{|x - x_0|^2}\right) \\
&\geq 0 && \text{for } |x - x_0| < \delta
\end{aligned}$$

Therefore

$$f(x) \geq f(x_0) \text{ for } |x - x_0| < \delta$$

Hence x_0 is a local minimum.

We can prove 2 the exact same way.

For 3, suppose that $\lambda,\ \lambda'$ are two eigenvalues of $D^2 f(x_0)$ with

$$\lambda > 0, \quad \lambda' < 0$$

Let $v,\ v'$ be the corresponding normalized eigenvectors. Take

$$x = x_0 + tv$$

Therefore

$$\begin{aligned} f(x_0+tv)-f(x_0) &= \frac{1}{2}\langle D^2 f(x_0)tv, tv\rangle + o(|t|^2) \\ &= \frac{1}{2}t^2\left(\langle D^2 f(x_0)v, v\rangle + \frac{o(|t|^2)}{t^2}\right) \\ &= \frac{1}{2}t^2\left(\lambda + o(|t|^2)/t^2\right) \\ &\geq 0 && t \text{ small enough} \end{aligned}$$

Similarly, for t small enough

$$f(x_0+tv')-f(x_0) \leq 0$$

Hence f has a saddle point at x_0. □

Example 2.42. Math 201:

- Solve $\nabla f(x_0) = 0$
- $D^2 f(x_0)$ find its eigenvalues.
- Apply second derivative test (if applicable).

2.5 Inverse function theorem.

Lemma 2.43 (Banach fixed point theorem)**.** *X complete metric space, $T\colon X \to X$ a contraction, i.e $\exists c \in [0,1]$ s.t*

$$d(f(x), f(y)) \leq cd(x,y)\ \forall x, y \in X$$

Then f admits a unique fixed point.

Proof. Take $x_1 \in X$ and $x_{n+1} = f(x_n)$, we show that x_n is Cauchy. Indeed,

$$d(x_2, x_3) = d(x_2, f(x_2)) \leq Cd(x_2, x_1)d(x_4, x_3) \qquad \leq Cd(x_3, x_2) \leq C^2 d(x_2, x_1) \vdots$$
$$d(x_{n+1}, x_n) \leq C^{n-1} d(x_2, x_1)$$

Pick any $m > n$ then

$$\begin{aligned} d(x_m, x_n) &\leq d(x_{n+1}, x_n) + d(x_{n+2}, x_{n+1}) + \cdots + d(x_{m-1}, x_m) \\ &\leq (C^{n-1} + C^n + \cdots + C^{m-2})d(x_2, x_1) \\ &= C^{n-1}(1 + C + C^2 + \cdots + C^{m-n-1})d(x_2, x_1) \\ &\leq \frac{C^{n-1}}{1-C} d(x_2, x_1) \end{aligned}$$

Let $\varepsilon > 0$ then

$$\frac{C^{n-1}}{1-C} d(x_2, x_1) \to 0$$

i.e $\exists N$ s.t $\forall n \geq N$ we have

$$\frac{C^{n-1}}{1-C} d(x_2, x_1) < \varepsilon$$

Therefore for $m, n \geq N$ we have

$$d(x_m, x_n) < \varepsilon$$

Hence x_n is Cauchy therefore it converges to some x_0 (since X is complete). Moreover, L is Lipschitz continuous, and $x_{n+1} = f(x_n)$ let $n \to \infty$ we get

$$x_0 = f(x_0)$$

we have found a fixed point.

Let x_0, x_0' be fixed points,

$$\begin{aligned} d(f(x_0), f(x_0')) &\leq Cd(x_0, x_0') \\ d(x_0, x_0') &< d(x_0, x_0') \end{aligned}$$

Therefore $d(x_0, x_0') = 0$ hence $x_0' = x_0$. □

Theorem 2.44 (Inverse function theorem). *$U \subset \mathbb{R}^n$ open and $f\colon U \to \mathbb{R}^n$ a C^1 function and $x_0 \in U$ if $\det[Df(x_0)] \neq 0$ then $\exists O_1$ open neighborhood of x_0 and O_2 open neighborhood of $f(x_0)$ s.t $f\colon O_1 \to O_2$ is bijective. Moreover, $f^{-1} \in C_1(O_2)$ and*

$$(Df^{-1})(f(x)) = (Df(x))^{-1} \quad \forall x \in O_1$$

We say that f is a local diffeomorphism.

Remark 2.45. It is enough to assume that $Df(x_0) = I_n$; suppose that we have the setting of the theorem

$$\det(Df(x_0)) \neq 0$$

Then take

$$g(x) = (Df(x_0))^{-1} f(x) = (Df(x_0)) \circ f(x)$$

Therefore by chain rule and the fact that $DT = T$, we have

$$Dg(x_0) = (Df(x_0))^{-1}(Df(x)) = I_n$$

Apply the IFT at g, we find $O_1 \ni x_0$ and $O_2 \ni g(x_0)$ s.t $g\colon O_1 \to O_2$ is a Diffeomorphism, but

$$g(x) = Df(x_0)^{-1} \circ f(x) \implies f(x) = Df(x_0)(g(x))\colon O_1 \to Df(x_0)O_2$$

Note that $Df(x_0)^{-1}$ is a linear transformation, hence continuous and contains $f(x_0)$. Moreover the composition of diffeomorphisms is a diffeomorphism.

More generally, suppose $h, h^{-1} \in C_1$ therefore

$$h^{-1} \circ h = \mathrm{id}$$

therefore by chain rule

$$Dh^{-1}(h(x)) \circ Dh(x) = I_n \implies Dh^{-1}(h(x)) = (Dh(x))^{-1}$$

Remark 2.46. We will use the L_∞ norm on $\mathbb{R}^n$ and on $M_{n\times n}(\mathbb{R})$. Clearly

$$\frac{1}{\sqrt{n}} \leq \|x\|_\infty \leq \|x\|_2 \leq \sqrt{n}\|x\|_\infty$$

Remark 2.47. $GL_n(\mathbb{R})$ is an open subset of $M_{n\times n}(\mathbb{R})$.

Proof of the Inverse Function Theorem. We have that $Df(x_0) = I_n$ and Df is continuous, i.e. given any ε_0 $\exists \delta > 0$ s.t. for $\|x - x_0\|_\infty < \delta$ ($x \in B(x_0, \delta)$) we have

$$\|Df(x) - I\|_\infty < \varepsilon$$

and $[Df(x_0)] \in GL_n(\mathbb{R})$ (since open). But notice that

$$Df(x) - I = D(f(x) - x)$$

Define the function

$$g(x) = f(x) - x$$

We have that for $\|x - x_0\|_\infty < \delta$ we have that

$$\|Dg(x)\|_\infty < \varepsilon_0$$

By MVT, for $1 \leq i \leq n$,

$$\begin{aligned}
\|g_i(x) - g_i(y)\| &= |\nabla g_i(c) \cdot (x - y)| \\
&\leq |\nabla g_i(c)||x - y| \\
&\leq \|\nabla g_i(c)\|_\infty \|x - y\|_\infty \sqrt{n}\sqrt{n} \\
&\leq \|Dg(c)_\infty\| \|x - y\|_\infty \cdot n \\
&\leq \varepsilon_0 n \|x - y\|_\infty
\end{aligned}$$

Then if we left $\varepsilon_0 = \frac{1}{2n}$ then by the above $\exists \delta > 0$ s.t for $x \in B(x_0, \delta)_\infty$ we have that

$$|g_i(x) - g_i(y)| < \frac{1}{2}\|x - y\|_\infty$$

this holds for every $1 \leq i \leq n$, therefore

$$\begin{aligned}
\|g(x) - g(y)\|_\infty &< 1/2\|x - y\|_\infty \\
\|f(x) - x - f(y) + y\|_\infty &\leq \|x - y\|_\infty \\
\|x - y\|_\infty - \|f(x) - f(y)\|_\infty \leq \|f(x) - f(y) - (x - y)\|_\infty &\leq \|x - y\|_\infty \\
\|f(x) - f(y)\|_\infty &\geq \frac{1}{2}\|x - y\|_\infty
\end{aligned}$$

Therefore $f(x) = f(y) \iff x = y$ therefore we showed that f is injective on $B(x_0, \delta)$.

Let $y \in \mathbb{R}^n$, we wan to find $x \in B(x_0, r)$ s.t

$$f(x) = y$$

i.e. we want to find an x s.t

$$f(x) - x = y - x \iff x = x - f(x) + y = y - g(x)$$

This is equivalent to saying that x is a fixed point of the map

$$x \to y - g(x)$$

Hence, for $y \in \mathbb{R}^n$ define

$$T_y \colon B(x_0, \delta) \to \mathbb{R}^n, \quad T_y(x) = x - f(x) + y = y - g(x)$$

Notice that

$$\begin{aligned} \|T_y x - T_y z\|_\infty &= \|g(x) - g(z)\|_\infty \\ &\leq \frac{1}{2}\|x - z\|_\infty \end{aligned}$$

Take $X = \overline{B(x_0, \delta/2)}$, we want

$$T_y X \subseteq X$$

Want

$$\|T_y x - x_0\| \leq \delta/2 \quad \forall x \in \overline{B(x_0, \delta/2)}$$

But we have that

$$\begin{aligned} \|T_y x - x_0\| &= \|y - g(x) + g(x_0) - f(x_0)\| \\ &\leq \|g(x) - g(x_0)\|_\infty + \|y - f(x_0)\|_\infty \\ &\leq \frac{1}{2}\|x - x_0\|_\infty + \|y - f(x_0)\| \\ &\leq \frac{\delta}{4} + \|y - f(x_0)\|_\infty \\ &< \delta && \text{if } \|y - f(x_0)\| < \delta/4 \end{aligned}$$

Therefore for $y \in B(f(x_0), \delta/4)$, we have

$$T_y(\overline{B(x_0, \delta/2)}) \subseteq B(x_0, \delta/2) \subseteq \overline{B(x_0, \delta/2)}$$

Therefore for such y, we have that T_y is a contraction. I.e. for such y there exists a unique fixed point $x \in B(x_0, \delta/2)$, i.e.

$$T_y x = x \implies f(x) = y$$

Therefore

$$O_1 = B(x_0, \delta/2), \quad O_2 = B(f(x_0), \delta/4)$$

Therefore

$$f\colon O_1 \to O_2 \text{ is bijective}$$

It remains to show that $f^{-1}\colon O_2 \to O_1$ is C^1.

Recall that we already have $\forall x, x' \in O_1$

$$\|f(x) - f(x')\|_\infty \geq \frac{1}{2}\|x - x'\|$$

Therefore $\forall y, y' \in O_2$ we have

$$\|f^{-1}(y) - f^{-1}(y')\| \leq 2\|y - y'\|$$

We want to show that f^{-1} is differentiable

$$\begin{aligned} k &= f(f^{-1}(y+k)) - f(f^{-1}(y)) \\ &= Df(f^{-1}(y))\left(f^{-1}(y+k) - f^{-1}(y)\right) + o(\|f^{-1}(y+k) - f^{-1}(y)\|) \end{aligned}$$

Since $Df(f^{-1}(y)) \in GL_n(\mathbb{R})$, then

$$f^{-1}(y+k) - f^{-1}(y) = (Df(f^{-1}(y)))^{-1}k + (Df(f^{-1}(y)))^{-1}o(\|f^{-1}(y+k) - f^{-1}(y)\|)$$

It remains to show that

$$(Df(f^{-1}(y)))^{-1}o(\|f^{-1}(y+k) - f^{-1}(y)\|) = o(\|k\|)$$

Indeed,

$$\frac{o(\|f^{-1}(y+k) - f^{-1}(y)\|)}{\|k\|} = \frac{o(\|f^{-1}(y+k) - f^{-1}(y)\|)}{\|f^{-1}(y+k) - f^{-1}(y)\|} \cdot \frac{\|f^{-1}(y+k) - f^{-1}(y)\|}{\|k\|}$$

Therefore

$$\frac{o(\|f^{-1}(y+k) - f^{-1}(y)\|)}{\|k\|} = \frac{o(\|f^{-1}(y+k) - f^{-1}(y)\|)}{\|f^{-1}(y+k) - f^{-1}(y)\|} \cdot 2$$

Hence

$$\lim_{k \to 0} \frac{o(\|f^{-1}(y+k) - f^{-1}(y)\|)}{\|k\|} = 0$$

Therefore by continuity of $Df(f^{-1}(y))^{-1}$ we get that

$$Df(f^{-1}(y))^{-1}\left(\frac{o(\|f^{-1}(y+k) - f^{-1}(y)\|)}{\|k\|}\right)$$

Therefore f^{-1} is differentiable and

$$Df^{-1}(y) = Df(f^{-1}(y))^{-1}$$

Moreover, since Df is continuous, the map $A \to A^{-1}$ is continuous and f^{-1} is continuous. Therefore Df^{-1} is continuous, being the composition of continuous functions. □

Corollary 2.48. *Say we have $U \subseteq \mathbb{R}^n$ and $f\colon U \to \mathbb{R}^n$ C^1 function, if $\det[Df(x)] \neq 0$ for every $x \in U$, then f is an open map.*

Proof. Homework. □

2.6 Implicit function theorem.

Motivation. $x, y \in \mathbb{R}$ defined so that

$$f(x, y) = 0$$

Suppose $y := y(x)$, i.e. for each x there exists a unique y s.t.

$$f(x, y) = 0$$

Suppose that $y(x)$ is C^1, what is $y'(x)$?

Notice that $f(x, y(x)) = 0\ \forall x$, therefore by chain rule we get

$$\frac{\partial f}{\partial x} + y'(x)\frac{\partial f}{\partial y} = 0$$

Therefore

$$y'(x) = -\frac{f_x(x, y(x))}{f_y(x, y(x))}$$

provided that $f_y \neq 0$,

More generally suppose that we have $f\colon \mathbb{R}^{n+m} \to \mathbb{R}^m$ and take (x, y) with $x \in \mathbb{R}^n,\ y \in \mathbb{R}^m$. Consider the level curve

$$f(x, y) = 0$$

suppose that for each x there exists some unique y s.t.

$$f(x, y) = 0$$

let h be that function and assign for each $x \in \mathbb{R}^n$ some $h(x)$ s.t.

$$f(x, h(x)) = 0, \quad h\colon R^n \to \mathbb{R}^m$$

Suppose that h is C^1, we want to find $Dh(x)$.

We define

$$g(x) := (x, h(x))$$

therefore notice that

$$f(x, h(x)) = f \circ g(x) = 0$$

Hence differentiating we get

$$\begin{aligned} D(f \circ g)(x) &= 0 \\ Df(x, h(x))\, Dg(x) &= 0 \end{aligned}$$

In matrix form

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} & \frac{\partial f_1}{\partial y_1} & \cdots & \frac{\partial f_1}{\partial y_m} \\ \vdots & & \vdots & \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} & \frac{\partial f_m}{\partial y_m} & \cdots & \frac{\partial f_m}{\partial y_m} \end{pmatrix} \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & 0 & \cdots & 1 \\ \frac{\partial h_1}{\partial x_1} & \cdots & & \frac{\partial h_1}{\partial x_n} \\ \vdots & & & \vdots \\ \frac{\partial h_m}{\partial x_1} & \cdots & & \frac{\partial h_m}{\partial x_n} \end{pmatrix} = 0$$

$$\begin{pmatrix} D_x f(x, h(x)) & D_y f(x, h(x)) \end{pmatrix} \begin{pmatrix} I_n \\ Dh(x) \end{pmatrix} = 0$$

$$(D_x f)(x, h(x)) + D_y f(x, h(x)) Dh(x) = 0$$

Hence provided that $D_y(x, h(x))$ is invertible we get that

$$Dh(x) = (D_y f(x, h(x)))^{-1} D_x f(x, h(x))$$

More motivation. $U \subset \mathbb{R}^2$ and $f(x, y) = 0$ we need to solve y as a function of x. say there exists $(x_0, y_0) \in U$ s.t. $f(x_0, y_0) = 0$. Say f is C^1, therefore

$$f(x, y) \sim f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

We want to solve y as a function of x when $f(x, y) = 0$. Looking only at the RHS, to solve y uniquely as a function of x we need

$$\begin{aligned} f_y(x_0, y_0)(y - y_0) &= -f_x(x - x_0) \\ y - y_0 &= \frac{f_x}{f_y}(x - x_0) \end{aligned}$$

provided that $f_y \neq 0$.

Theorem 2.49 (Implicit Function Theorem). *$U \subset \mathbb{R}^{n+m}$ and $f \colon U \to \mathbb{R}^m$ a C^1 function, $(x_0, y_0) \in U$ then say $f(x_0, y_0) = 0$. Write $(x, y)\colon\ x \in \mathbb{R}^n,\ y \in \mathbb{R}^m$. Assume that $\det[D_y f(x_0, y_0)] \neq 0$, then $\exists U_0 \ni x_0$ and $V_0 \ni y_0$ open neighborhoods, s.t. for every $x \in U_0$ there exists a unique $y \in V_0$ s.t. $f(x, y) = 0$. Moreover, the function*

$$h \colon U_0 \to V_0$$

s.t. $f(x, h(x)) = 0$ *is a* C^1 *function.*

Remark 2.50. In the scenario above, by chain rule we get that

$$Dh(x) = -(D_y f(x, h(x)))^{-1} \circ D_x f(x, h(x))$$

Usually used at x_0,

$$Dh(x_0) = -(D_y f(x_0, h(y_0)))^{-1} \circ D_x f(x_0, h(x_0))$$

Proof. Take $F\colon U \to \mathbb{R}^{n+m}$ with

$$F(x, y) = (x, f(x, y)) = (x_1, \cdots, x_n, f_1, \cdots, x_m)$$

notice that $F(x_0, y_0) = (x_0, f(x_0, y_0)) = (x_0, 0)$, and that

$$DF(x_0, y_0) = \begin{pmatrix} I_n & 0 \\ D_x f & D_y F \end{pmatrix}$$

Therefore

$$\det(DF(x_0, y_0)) = \det(D_y f(x_0, y_0)) \neq 0$$

then by the inverse function theorem, there exists $U_0 \times V_0 \ni (x_0, y_0)$ an open subset with $U_0 \ni x_0$ and $V_0 \ni y_0$ and $W_0 \ni (x_0, 0)$ an open neighborhood s.t.

$$F\colon U_0 \times V_0 \to W_0$$

a diffeomorphism.

Now take $(x', y') \in W_0$ then there exists an unique $(x, y) \in U_0 \times V_0$ s.t.

$$F(x, y) = (x', y') = (x, f(x, y))$$

i.e. there exists a unique $y \in V_0$ s.t. $f(x, y) = y'$.

Therefore we can choose U_0 small enough so that $U_0 \times \{0\} \subseteq W_0$. Given $x \in U_0$ we have $(x, 0) \in W_0$, then $\exists\, ! y \in V_0$ s.t. $f(x, y) = 0$. This defines a function $h\colon U_0 \to V_0$ s.t. $f(x, h(x)) = 0$. We only need to show that h is C^1, indeed

$$\begin{aligned} F(x, h(x)) &= (x, f(x, h(x))) \\ &= (x, 0) \end{aligned}$$

Therefore

$$F^{-1}(x, 0) = (x, h(x))$$

but by the inverse function theorem, we have that F^{-1} is C^1 hence h is C^1. □

Corollary 2.51 (Lagrange Multiplier)**.** *Let* $g\colon \mathbb{R}^n \to \mathbb{R}$ *function and take the level set*

$$S = \{x \in \mathbb{R}^n \colon g(x) = 0\}$$

Let $U \supset S$ be an open subset of $\mathbb{R}^n$ s.t. $g \in C^1(U)$ and $\nabla g(x) \neq 0 \; \forall x \in S$. Let $f\colon U \to \mathbb{R}$ be differentiable, if f attains an local minimum (maximum) on S at some point x_0, then $\nabla f(x_0) = \lambda \nabla g(x_0)$ for some $\lambda \in \mathbb{R}$.

Proof. We have that $\nabla g(x_0) \neq 0$ therefore $\frac{\partial g}{\partial x_i}(x_0) \neq 0$ for some i, wlog say that

$$\frac{\partial g}{\partial x_n}(x_0) \neq 0$$

We can write

$$g\colon \mathbb{R}^{n-1+1} \to \mathbb{R}, \quad x = (x', x_n)$$

With $\frac{\partial g}{\partial x_n}(x_0', (x_0)_n) \neq 0$ then by the implicit function theorem applied to $n \to n-1$ and $m \to 1$, there exists a neighborhood $U_0 \ni x_0'$ and some $h\colon U_0 \to \mathbb{R}$ s.t. $g(x', h(x')) = 0$. Differentiating w.r.t. x' we get

$$D_{x'} g + \frac{\partial g}{\partial x_n} Dh(x') = 0$$

Hence

$$\nabla h(x') = -\frac{\nabla_{x'} g}{g_{x_n}}$$

i.e.

$$\nabla h(x_0') = -\frac{1}{g_{x_n}(x_0)} \nabla_{x'} g(x_0)$$

Define

$$F\colon U_0 \to \mathbb{R}, \quad F(x') = f(x', h(x'))$$

But notice that $(x', h(x')) \in S$ but F admits a local minimum (or maximum) at x_0'; therefore

$$\nabla F(x_0') = 0 = \nabla_{x'} f(x_0) + \frac{\partial f}{\partial x_n}(x_0) \nabla_{x'} h(x_0')$$

Hence

$$\nabla_{x'} f(x_0) = \frac{f_{x_n}(x_0)}{g_{x_n}(x_0)} \nabla_{x'} g(x_0)$$

But also (trivially)

$$\frac{\partial f}{\partial x_n}(x_0) = \frac{f_{x_n}(x_0)}{g_{x_n}(x_0)} g_{x_n}(x_0)$$

Therefore letting $\lambda = \frac{f_{x_n}(x_0)}{g_{x_n}(x_0)}$ we get

$$\nabla_x f(x_0) = \lambda \nabla g(x_0)$$

$\square$

Example 2.52. Consider the following system

$$\begin{cases} x_1 e^{x_2} + x_3 + \sin x_4 + x_5^2 = 1 \\ x_1 \sin x_4 + x_2 + x_4 - x_3 x_5 = 0 \end{cases}$$

Verify that $(1,0,0,0,0)$ solves the system of equations; show that in a neighborhood of $(0,0,0)$ there exists a C^1 function $g\colon (g_1, g_2)$ s.t. $g(0,0,0) = (1,0)$ and for every (x_3, x_4, x_5) in tht neighborhood we have that

$$(g_1(x_3, x_4, x_5), g_2(x_3, x_4, x_5), x_3, x_4, x_5)$$

are the unique solutions to the system.

Then calculate $\frac{\partial g_i}{\partial x_i}$ with $i = 1, 2$ $j = 3, 4, 5$.

Solution. $f\colon \mathbb{R}^{3+2} \to \mathbb{R}^2$ by

$$f(x_1, x_2, x_3, x_4, x_5) = (x_1 e^{x_2} + x_3 + \sin x_4 + x_5^2 - 1, x_1 \sin x_4 + x_2 + x_4 - x_3 x_5 = 0)$$

Note that

$$f(1,0,0,0,0) = (0,0)$$

Our objective is to solve

$$f(x_1, x_2, x_3, x_4, x_5) = (0,0)$$

Note that

$$Df(1,0,0,0,0) = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 \end{pmatrix}$$

Hence

$$\det(D_{x_1,x_2} f(1,0,0,0,0)) = \det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = 1 \neq 0$$

Then by the implicit function theorem, there exists a neighborhood U of $(0,0,0)$ and a C^1 function $g\colon U \to \mathbb{R}^2 = (g_1, g_2)$ s.t.

$$f(g(x_3, x_4, x_5), g_2(x_3, x_4, x_5), x_3, x_4, x_5) = (0,0,0)$$

for all $(x_3, x_4, x_5) \in U_0$. Note

$$Dg(0,0,0) = -(D_{x_1,x_2} f(1,0,0,0,0))^{-1} D_{x_3,x_4,x_5} f(1,0,0,0,0)$$

In matrix form

$$\begin{aligned}\begin{pmatrix} \frac{\partial g_1}{\partial x_3}(0,0,0) & \frac{\partial g_1}{\partial x_4}(0,0,0) & \frac{\partial g_1}{\partial x_5}(0,0,0) \\ \frac{\partial g_2}{\partial x_3}(0,0,0) & \frac{\partial g_2}{\partial x_4}(0,0,0) & \frac{\partial g_2}{\partial x_5}(0,0,0) \end{pmatrix} &= -\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \end{pmatrix} \\ &= -\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -1 & 1 & 0 \\ 0 & -2 & 0 \end{pmatrix}\end{aligned}$$

3 Riemann Integration

Convention. A rectangle $R \subset \mathbb{R}^n$ has the form

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$$

We define the volume of R

$$\mathrm{vol}(R) := (b_1 - a_1) \cdots (b_n - a_n)$$

Definition 3.1. $[a, b] \subseteq \mathbb{R}$ a partition P of $[a, b]$ is an ordered set

$$P = \{t_0, \cdots, t_k\}$$

s..t

$$t_0 = a < t_1 < t_2 < \cdots < t_k = b$$

Given a partitions $P_1, \cdots, P_n$ of $[a_1, b_1], \cdots, [a_n, b_n]$, we say that

$$P = \{P_1, \cdots, P_n\}$$

is a partition of the rectangle $R = [a_1, b_1] \times \cdots \times [a_n, b_n]$ formed of all sub-rectangles with end-points in P_i. A partition $P = \{P_1, \cdots, P_n\}$ decompose R into $R_1, \cdots, R_N$ with non-overlapping interiors, with

$$\mathrm{vol}(R) := \sum_{i=1}^{N} \mathrm{vol}(R_i)$$

$A \subseteq \mathbb{R}$ a rectangle and $f \colon A \to \mathbb{R}$ bounded:

$$m \leq f \leq M$$

Let P be a partition of A with corresponding rectangles $R_1, \cdots, R_n$. Let $m_i = \inf\{f(x) \colon x \in R_i\}$ and $M_i = \sup\{f(x) \colon x \in R_i\}$. Then we define the upper and lower sums

$$L(P, f) = \sum_{i=1}^{N} m_i \,\mathrm{vol}(R_i), \quad V(P, f) = \sum_{i=1}^{N} M_i \,\mathrm{vol}(R_i)$$

Since $m \leq m_i \leq M_i \leq M$ we have that

$$m \operatorname{vol}(A) \leq L(P, f) \leq V(P, f) \leq M \operatorname{vol}(A)$$

Definition 3.2. Given a partition P of $[a, b]$ a refinement P^* of P is a partition of $[a, b]$ s.t. $P \subset P^*$.

Given a partition $P = \{P_1, \cdots, P_n\}$ of a rectangle $A = [a_1, b_1] \times \cdots \times [a_n, b_n]$; a refinement P^* of P is a partition of A

$$P^* = \{P_1^*, \cdots, P_n^*\}$$

s.t. each P_i^* is a refinement of P_i.

Given $P = \{P_1, \cdots, P_n\}$ and $Q = \{Q_1, \cdots, Q_n\}$ are partitions of a rectangle A, then we define

$$P \cup Q := \{P_1 \cup Q_1, \cdots, P_n \cup Q_n\}$$

to be the common refinement of P and Q.

Proposition 3.3. *Given a rectangle A and $f\colon A \to \mathbb{R}$ bounded, and P partition of A and P^* refinement of P, then*

$$L(P^*, f) \geq L(P, f) \text{ and } U(P^*, f) \leq U(P, f)$$

Proof. P^* a refinement of P and let R be a rectangle obtained from P, then

$$R = R_1^* \cup \cdots \cup R_k^*$$

with R_i^*'s rectangles with disjoint interiors corresponding to P^*; Notice that

$$\begin{aligned} L(P, f) &= \sum_{R \in P} \inf \{f(x) \colon x \in R\} \operatorname{vol}(R) \\ &= \sum_{R \in P} \sum_{i=1}^{k_R} \inf \{f(x) \colon x \in R\} \operatorname{vol}(R_i^*) \\ &\leq \sum_{R \in P} \sum_{i=1}^{k_R} \inf \{f(x) \colon x \in R^*\} \operatorname{vol}(R_i^*) \\ &= L(P^*, f) \end{aligned}$$

similarly for $U(P, f) \geq U(P^*, f)$. □

Corollary 3.4. *Given $A \subseteq \mathbb{R}^n$ a rectangle, $f\colon A \to \mathbb{R}$ bounded and $P_1,\ P_2$ partition of A then*

$$L(P_1, f) \leq U(P_2, f)$$

Hence

$$\sup_P L(P, f) \leq \inf_P U(P, f)$$

Proof. take a common refinement. □

Definition 3.5. Let $f\colon A \to \mathbb{R}$ bounded with A a rectangle, we say that f is integrable over $\mathbb{R}$ iff

$$\sup L(P, f) = \inf U(P, f)$$

and we write

$$\int_A f dx := \sup_P L(P, f) = \inf_P U(P, f)$$

More generally we denote

$$\underline{\int_A} f dx = \sup L(P, f), \quad \overline{\int_A} f dx = \inf L(P, f)$$

Note that these always exist with

$$\underline{\int_A} f dx \leq \overline{\int_a} f dx$$

with equality iff f is integrable, in this case we write $f \in \mathcal{R}(A)$.

Example 3.6. A is a rectangle and $f\colon A \to \mathbb{R}$ s.t.

$$f(x) = c, \quad \forall x \in A$$

Then f is integrable and

$$\int_A c dx = c \operatorname{vol} A$$

in particular, if $c = 1$ we have that

$$\int dx = c \operatorname{vol} A$$

Proof. Let P be a partition of A into rectangles $R_1, \cdots, R_N$, notice that

$$m_i = \inf \{f(x) \colon x \in R_i\} \implies L(P, f) = \sum_{i=1}^{n} c \operatorname{vol}(R_i) = c \operatorname{vol}(A) \; \forall P$$

Therefore

$$\sup L(P, f) = c \operatorname{vol} A$$

Similarly we have that

$$M_i = c \implies U(P, f) = c \operatorname{vol} A \; \forall P$$

Hence

$$\inf U(P, f) = c \operatorname{vol} A$$

Therefore f is integrable and

$$\int_A c dx = c \operatorname{vol}(A)$$

Example 3.7. Let A be a rectangle $f(x) = \begin{cases} 1 & x_1 \in \mathbb{Q} \\ 0 & x_1 \notin \mathbb{Q} \end{cases}$, f is not integrable.

Proof. Let P be a partition of A into rectangles $R_1, \cdots, R_N$. Therefore

$$m_i = 0 \implies L(P, f) = 0 \implies \sup_P L(P, f) = 0$$

and

$$M_i = 1 \implies U(P, f) = \operatorname{vol}(A) \implies \inf U(P, f) = \operatorname{vol}(A)$$

Therefore $f \notin \mathcal{R}$.

Proposition 3.8 (Integrability criterion). *$f\colon A \to \mathbb{R}$ bounded and A a rectangle; then $f \in \mathcal{R}(A)$ iff $\forall \varepsilon > 0\ \exists P$ a partition of A s.t. $U(P, f) - L(P, f) < \varepsilon$.*

Proof. $\implies$ Assume that f is integrable and let $\varepsilon > 0$ then $\int_A f dx = \sup L(P, f)$ therefore $\exists P_1$ partition s.t.

$$\int_A f dx - \varepsilon < L(P_1, f) \leq \int_A f dx$$

Similarly $\exists P_2$ s.t.

$$\int_A f dx \leq U(P_2, f) < \int_A f dx + \varepsilon/2$$

Let $P = P_1 \cup P_2$ be the common refinement of P_1 and P_2 then

$$U(P, f) \leq U(P_2, f) < \int_A f dx + \varepsilon/2$$

and

$$L(P, f) \geq L(P_1, f) > \int_A f dx - \varepsilon$$

Hence

$$U(P, f) - L(P, f) < \varepsilon/2$$

$\impliedby$ Take f that respects the integrability criterion, given $\varepsilon > 0\ \exists P_\varepsilon$ s.t.

$$U(P_\varepsilon, f) - L(P_\varepsilon, f) < \varepsilon$$

Therefore

$$0 \geq \inf_P U(P, f) - \sup_P L(P, f) < \varepsilon$$

Take $\varepsilon \to 0$ we get

$$\inf_P U(p, f) - \sup_P L(P, f) = 0$$

□

Corollary 3.9. *$A \subseteq \mathbb{R}^n$ a rectangle and $f\colon A \to \mathbb{R}$, if f is continuous then f in integrable.*

Proof. f is continuous and A is compact therefore f is bounded and moreover f is uniformly continuous. Let $\varepsilon > 0$, then $\exists \delta > 0$ s.t.

$$|f(x) - f(y)| < \frac{\varepsilon}{\mathrm{vol}(A) + 1}, \quad \forall |x - y| \delta$$

Let P be a partition of A into rectangles $R_1, \cdots, R_N$ with $\operatorname{diam} R_i < \delta$. Therefore for $x, y \in R_i$ we have that $|x - y| < \delta$ and therefore $|f(x) - f(y)| < \frac{\varepsilon}{\mathrm{vol}(A)+1}$. Therefore

$$\sup_{R_i} f - \inf_{R_i} f \leq \frac{\varepsilon}{\mathrm{vol}(A) + 1}$$

Therefore

$$\begin{aligned} U(P,f) - L(P,f) &= \sum_{i=1}^{N} (\sup_{R_i} f - \inf_{R_i}) \operatorname{vol}(R_i) \\ &\leq \frac{\varepsilon}{\operatorname{vol} A + 1} \sum_i \operatorname{vol}(R_i) \\ &< \varepsilon \end{aligned}$$

□

Remark 3.10. The converse is not true; let $A = [1,5] \times [1,3]$ and

$$f(x,y) = \begin{cases} 1 & 1 \leq x \leq 3 \\ 2 & 3 < x \leq 5 \end{cases}$$

Note that f is discontinuous $(3,y)$ with $1 \leq y \leq 3$. Let $2 > \varepsilon > 0$ and Take $P_1 = \{1, 3, 3+\varepsilon, 5\}$ and $P_2 = \{1, 3\}$; we get R_1, R_2, R_3 (draw the picture). We have that

$$\begin{aligned} m_1 &= 1 & M_1 &= 2 \\ m_2 &= 1 & M_2 &= 2 \\ m_3 &= 2 & M_3 &= 2 \end{aligned}$$

Therefore

$$U(P,f) - L(P,f) = 2\varepsilon$$

Hence f is integrable.

Proposition 3.11. *Let $A \subset \mathbb{R}^n$ be a rectangle and $f, g \colon A \to \mathbb{R}$ be integrable.*

1. *if $f \leq g$ on A then $\int_A f dx \leq \int_A g dx$.*
2. *$f + \alpha g$ is integrable $\forall \alpha \in \mathbb{R}$ with*

$$\int_A f + \alpha g dx = \int_A f dx + \alpha \int_A g dx$$

Proof. Let P be a partition of A into $R_1, \cdots, R_N$ on R_i but

$$f(x) \leq g(x) \; \forall x \in R_i$$

therefore

$$\sup_{R_i} f(x) \leq \sup_{R_i} g(x)$$

Hence

$$U(P,f) \leq U(P,g)$$

and

$$\int_A f dx \leq U(P,g), \quad \forall p$$

Therefore

$$\int_A f dx \leq \inf_P U(P, g) = \int_A g dx$$

We will show it for $\alpha = 1$, exercise show that

$$\int_A cf = c \int_A f$$

i.e. We will show that $f + g$ is integrable and

$$\int_A f + g dx = \int_A f dx + \int_B g dx$$

Let P_1, P_2 be partitions of A and let P be the common refinement of P_1, P_2. Then

$$\begin{aligned} U(P, f+g) &= \sum_{i=1}^{n} \sup \{f+g\} \operatorname{vol}(R_i) \\ &\leq \sum_{i=1}^{N} \sup f (\operatorname{vol} R_i) + \sum_{i=1}^{N} (\sup g) \operatorname{vol}(R_i) \\ &= U(P, f) + U(P, g) \\ &\leq U(P_1, f) + U(P_2, g) \end{aligned}$$

Hence we have $\forall P_1, P_2$

$$\inf_P U(P, f+g) \leq U(P_1, f) + U(P_2, g)$$

Take $\inf_{P_1} \inf_{P_2}$ and using that f, g are integrable we get that

$$\inf_P U(P, f+g) \leq \int_A f + \int_A f$$

Similarly

$$\sup_P L(P, f+g) \geq \int_A f + \int_A f$$

But

$$\sup L(P, f+g) \leq \inf(P, f+g)$$

hence

$$\inf L(P, f+g) = \sup U(P, f+g) = \int_A f dx + \int_A g dx$$

□

Definition 3.12. Let $E \subseteq \mathbb{R}^n$, we define the characteristic function

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

Theorem 3.13. *Let $A \subseteq \mathbb{R}^n$ a rectangle and $R \subseteq A$ also a rectangle. Then $\chi_R(x),\ \chi_{\mathring{R}}(x)$ are integrable and*

$$\int_A \chi_R(x) dx = \int_A = \chi_{\mathring{R}}(x) dx = \operatorname{vol}(R)$$

Proof. If $R = A$, we are done. We will show it for $f(x) = \chi_R(x)$, do the interior as an exercise. Given $\varepsilon > 0$, let R' be a rectangle s.t. $\exists U$ open set of A with

$$R' \supset U \supset R$$

Use the vertices of $R,\ R'$ and A to construct a partition P of A into rectangles $R_1, R_2, \cdots, R_N$; we take cases:

- **Case** $R_i = R$**.** Then $M_i = 1,\ m_i = 1$.
- **Case** $R_i \cap R = \varnothing$**.** Then $M_i = 0,\ m_i = 0$; in this case $R_i \subset A \setminus (\mathring{R}')$.
- **Case** $R_i \cap R \neq \varnothing$ **and** $R_i \neq R$**.** In this case $R_i \subset R' \setminus \mathring{R}$. $M_i = 1,\ m_i = 0$.

Therefore

$$U(P, f) - L(P, f) = \sum_{R \in \text{Case } 3} \text{vol}(R_i) = \text{vol}(R') - \text{vol}(R) < \varepsilon$$

Hence f is integrable, with

$$L(P, f) = \text{vol}(R)$$

and

$$U(P, f) < \text{vol}\, R + \varepsilon$$

Therefore $\forall \varepsilon > 0$ we have

$$\text{vol}(R) \leq \int_A \chi_R(x) dx < \text{vol}\, R + \varepsilon$$

hence

$$\int_A \chi_R(x) dx = \text{vol}(R)$$

□

Example 3.14. Let $A = [1, 5] \times [1, 3]$ and

$$f(x, y) = \begin{cases} 1 & 1 \leq x \leq 3 \\ 0 & 3 < x < 5 \end{cases}$$

Therefore

$$f(x, y) = \chi_{[1,3] \times [1,3]}(x)$$

Hence f is integrable and therefore

$$\int_A f(x, y) d(x, y) = 4$$

OTOH if

$$f(x, y) = \begin{cases} 1 & 1 \leq x < 3 \\ 0 & 3 \leq x \leq 5 \end{cases}$$

Notice that

$$f(x, y) = 1 - \chi_{[3,5] \times [1,3]}(x, y)$$

Therefore f is integrable and

$$\begin{aligned}\int_A f(x,y)d(x,y) &= \int_A 1 - \chi_{[3,5]\times[1,3]}(x,y)d(x,y)\\ &= \mathrm{vol}(A) - \mathrm{vol}([3,5]\times[1,3])\\ &= 8-4\\ &= 4\end{aligned}$$

Example 3.15. Let $A = [1,5]\times[1,3]$ and

$$f(x,y) = \begin{cases} 1 & 1 \le x \le 3 \\ 2 & 3 < x \le 5 \end{cases}$$

Notice that

$$f(x,y) = \chi_{[1,3]\times[1,3]} + 2\chi_{(3,5]\times[1,3]}$$

since we got $chi_{(3,5]\times[1,3]} \in R$ above, therefore f is integrable and

$$\begin{aligned}\int_A f(x,y)d(x,y) &= 4 + 2\cdot 8 + 2\cdot 4\\ &= 12\end{aligned}$$

Remark 3.16. if $A \subseteq B$ then $\chi_A(x) \le \chi_B(x)$ and $\chi_{A\cup B}(x) \le \chi_A(x) + \chi_B(x)$.

Proof. Trivial. □

Corollary 3.17. *Let $R_1, \cdots, R_N$ be a cover of rectangles of A, i.e.*

$$A \subseteq R_1 \cup \cdots \cup R_N$$

Then

$$\mathrm{vol}(A) \le \mathrm{vol}(R_1) + \cdots + \mathrm{vol}(R_N)$$

3.1 Measure zero sets.

Definition 3.18. Let $E \subset \mathbb{R}^n$, we say that E has measure 0 iff for every $\varepsilon > 0$, there exists a countable cover of rectangles $\{R_1, R_2, \cdots\}$ of the set E s.t.

$$\sum_{i=1}^{\infty} \mathrm{vol}(R_i) < \varepsilon$$

If $\forall\varepsilon$ we can find a finite set we say that E has content zero.

Example 3.19. Let $x = (x_1, \cdots, x_n) \in \mathbb{R}^n$, then $E = \{x\}$ has measure 0. Indeed, let $\varepsilon > 0$ then

$$R = [x_1 - \varepsilon/4, x_1 + \varepsilon/4] \times [x_2 - 1/2, x_2 + 1/2] \times \cdots \times [x_n - 1/2, x_n + 1/2]$$

Note that $E \subset R_1$ with $\mathrm{vol}(R_1) = \varepsilon/2$.

Example 3.20. If A has measure zero and $B \subset A$ had measure zero.

Proposition 3.21. *Any countable union of sets of measure zero has measure zero.*

Proof. Let $E_1, E_2, \cdots$ have measure zero and define

$$E = \bigcup_{i=1}^{\infty} E_i$$

Let ε, since E_1 has measure we can cover E_1 with $R_1^1, R_2^1, \cdots$ s.t.

$$\sum_{i=1}^{N} \mathrm{vol}(R_i^1) < \varepsilon/2$$

Similarly for E_2 we can cover it by $R_1^2, R_2^2, \cdots$ s.t.

$$\sum_{i=1}^{\infty} \mathrm{vol}(R_i^2) < \varepsilon/2^2$$

In general E_k has measure zero then we can cover E_k with rectangles

$$R_1^k, R_2^k, \cdots$$

with

$$\sum_{i=1}^{\infty} \mathrm{vol}(R_i^k) < \varepsilon/2^k$$

Then we can cover E by

$$\bigcup_{k=1}^{\infty} \bigcup_{i=1}^{\infty} R_i^k$$

with

$$\sum_{k=1} \sum_{i=1}^{\infty} \mathrm{vol}(R_i^k) < \sum_{k=1}^{\infty} \varepsilon/2^k = \varepsilon$$

□

Corollary 3.22. *Any countable set in $\mathbb{R}^n$ has measure zero.*

Example 3.23. A rectangle $A \subset \mathbb{R}^n$ does not have measure zero.

Let $\mathring{R}_1, \mathring{R}_2, \cdots$ be a cover of A. But A is compact; then $\exists i_1, \cdots, i_n$ s.t.

$$A \subseteq R_{i_1} \cup \cdots \cup R_{i_n}$$

Hence

$$\mathrm{vol}(A) \leq \sum_{k=1}^{n} R_{i_k}$$

Therefore for $0 < \varepsilon < \mathrm{vol}(A)$ there is no (countable) cover of rectangles s.t.

$$\sum_{i=1}^{\infty} \mathrm{vol}(R_i) < \varepsilon$$

Consequence. If $\mathring{E} \neq \varnothing$, then E does not have measure zero.

Remark 3.24. The converse of the above is not true: i.e. if $\mathbb{E}$ is empty, we could still have that E doesn't have measure zero.

For example $\mathbb{R} \setminus \mathbb{Q}$ does not have measure zero in $\mathbb{R}$. In fact, suppose that it does, then $\mathbb{R} = (\mathbb{R} \setminus \mathbb{Q}) \cup \mathbb{Q}$ has measure zero, contradiction since $\mathbb{R}$ is open.

Remark 3.25. We must be careful of the ambient space: $\mathbb{R}$ has measure zero in $\mathbb{R}^2$ (more generally $\mathbb{R}^n$ has measure zero in $\mathbb{R}^{n+k}$ for $k > 0$).

We will show that $(0, \infty) \times \{0\}$ has measure 0 in $\mathbb{R}^2$ similarly $(-\infty, 0) \times \{0\}$ will have measure in $\mathbb{R}^2$. Fix $\varepsilon > 0$ let

$$R_1 = [0,1] \times [-\frac{\varepsilon/4}{2}, \frac{\varepsilon/4}{2}],\ R_2 = [0,1] \times [\frac{\varepsilon/4}{2^2}, \frac{\varepsilon/4}{2^2}], \cdots,\ \mathbb{R}_n = [n-1, n] \times [-\frac{\varepsilon/4}{2_n}, \frac{\varepsilon/4}{2^n}]$$

Hence

$$\mathrm{vol}(R_1) = \varepsilon/4,\ \mathrm{vol}(R_2) = \frac{\varepsilon/4}{2}, \cdots,\ \mathrm{vol}(R_n) = \frac{\varepsilon/4}{2^{n-1}}$$

Notice that $\{R_1, R_2, \cdots\}$ covers $[0, \infty)$ hence

$$\sum_{i=1}^{\infty} = \varepsilon/4 \cdot 2 < \varepsilon$$

Another way to do this is to show that $[-n, n] \times \{0\}$ has measure 0 Hence $\mathbb{R} = \bigcup_{n \in \mathbb{N}} [-n, n] \times \{0\}$ has measure zero.

Definition 3.26. Given $E \subseteq \mathbb{R}^n$, the boundary of E is defined as follows: $x \in \partial E$ iff $\forall \varepsilon > 0,\ N_\varepsilon(x) \cap E \neq \varnothing$ and $N_\varepsilon(x) \neq \varnothing$. i.e.

$$\partial E = \overline{E} \setminus \mathring{E}$$

Example 3.27. If $A = [a_1, b_1] \times \cdots \times [a_n, b_n]$ then

$$\begin{aligned} \partial A = \{a_1\} &\times [a_2, b_2] \times \cdots \times [a_n, b_n] \\ &\cup \{b_1\} \times [a_2, b_2] \times \cdots \times [a_n, b_n] \\ \vdots \\ &\cup [a_1, b_1] \times \cdots \times \{b_n\} \end{aligned}$$

therefore ∂A has measure zero.

Note that this is special to rectangles, indeed $\partial \mathbb{Q} = \mathbb{R}$ which doesn't have measure zero in $\mathbb{R}$.

3.2 Lebesgue Criterion for Riemann integrability.

Definition 3.28 (Oscillation function)**.** Let X, Y be metric spaces and $f\colon X \to Y$, take $E \subseteq X$ then we define the oscillation of f on E to be

$$\operatorname{osc}_f(E) = \sup_{x_1, x_2 \in E} d_Y(f(x_1), f(x_2))$$

For $x \in X$ we define the oscillation of f at x

$$\omega_f(x) = \inf_{\delta > 0} \operatorname{osc}_f(N_\delta(x)) = \lim_{\delta \to 0 \operatorname{osc}(N_\delta(x))}$$

i.e.

$$\omega_f(x) = \inf_{\delta > 0} \left(\sup_{x_1, x_2 \in N_\delta(x)} d_y(f(x_1), f(x_2)) \right)$$

Proposition 3.29. *f continuous at $x_0 \iff \omega_f(x_0) = 0$.*

Corollary 3.30. *Given a function f the set of discontinuities*

$$D_f = \{x \in X \colon \omega_f(x) > 0\}$$

Proof. $\implies$ Let $\varepsilon > 0$ f is continuous at x_0, then $\exists \delta_0 > 0$ s.t.

$$d(f(x), f(x_0)) < \varepsilon/2 \; \forall x \in N_\delta(x_0)$$

So $x_1, x_2 \in N_\delta(x_0)$ therefore

$$d_Y(f(x_1), f(x_2)) < \varepsilon$$

Therefore

$$\operatorname{osc}(N_{\delta_0}(x_0)) \leq \varepsilon$$

Hence

$$\omega_f(x_0) = \inf_{\delta > 0}(\operatorname{osc}(N_\delta(x_0))) \leq \varepsilon$$

Hence

$$\omega_f(x_0) = 0$$

$\impliedby$ Fix $\varepsilon > 0$ suppose $\omega_f(x_0) = \inf_{\delta > 0}(\operatorname{osc} N_\delta(x_0)) = 0$. Therefore $\exists \delta_0 > 0$ s.t.

$$\operatorname{osc}(N_{\delta_0}(x_0)) = \sup_{x_1, x_2 \in N_{\delta_0}(x_0)} < \varepsilon$$

Therefore $\forall x \in N_{\delta_0}$ we have

$$d(f(x), f(x_0)) < \varepsilon$$

hence f is continuous at x_0.

$\square$

Proposition 3.31. *Let $f\colon X \to Y$ be a (not necc. continuous) map, and let*

$$G = \{x \in X : \omega_f(x) < \alpha\}$$

then G is open $\forall \alpha > 0$.

We say that ω is (lower) semi-continuous.

Proof. Take $x_0 \in G$, then

$$\omega_f(x_0) = \inf_\delta \operatorname{osc}(N_\delta(x_0)) < \alpha$$

Therefore $\exists \delta_0 > 0$ s.t.

$$\sup_{x_1, x_2 \in N_{\delta_0}(x_0)} (d_Y(f(x_1), f(x_2))) < \alpha$$

We claim that

$$N_{\delta_0}(x_0) \subset G$$

Indeed take $x \in N_{\delta_0}(x_0)$, then $\exists r > 0$ s.t.

$$N_r(x) \subset N_{\delta_0}(x_0)$$

Hence

$$\operatorname{osc}(N_r(x)) \leq \operatorname{osc}(N_{\delta_0}(x_0)) < \alpha$$

Therefore

$$\omega_f(x) = \inf_{\delta > 0} \operatorname{osc}(\mathrm{N}_\delta(\mathrm{x})) \leq \operatorname{osc}(N_r(x)) < \alpha$$

Hence $x \in G$. Hence x_0 is an interior point and so G is open. □

Proposition 3.32. *Let A a rectangle in $\mathbb{R}^n$ and $f\colon A \to \mathbb{R}$ suppose that for every $x \in A$ we have*

$$\omega_f(x) < \alpha$$

for some α, then there exists some partition P_α of A s.t.

$$U(P_\alpha, f) - L(P_\alpha, f) \leq \alpha \operatorname{vol}(A)$$

Proof. $\omega_f(x) = \inf_\delta \sup_{x_1, x_2 \in N_\delta(x)} |f(x_1) - f(x_2)| < \alpha$ then there exists a rectangle R_X s.t.

$$\sup_{x_1, x_2 \in R_X} |f(x_1) - f(x_2)| < \alpha$$

Therefore

$$0 \leq \sup_{R_x} F - \inf_{R_X} \leq \alpha$$

Now since A is compact we can find some finite index set $x_1, \cdots, x_k$ s.t.

$$A \subseteq \mathring{R}_{x_1} \cup \cdots \cup \cdots \mathring{R}_{x_k} \subseteq R_{x_1} \cup \cdots \cup \cdots R_{x_k}$$

Use the vertices and edges of R_{x_i}'s and A to construct a partition $P_\alpha = \left\{\tilde{R}_1, \tilde{R}_2, \cdots, \tilde{R}_N\right\}$ of A.

But notice

$$\tilde{R}_i \subseteq R_{x_t} \text{ for some } t$$

Hence

$$\sup_{\tilde{R}_i} f - \inf_{\tilde{R}_i} f \leq \alpha$$

Hence

$$\begin{aligned} U(P_\alpha, f) - L(P_\alpha, f) &= \sum_{i=1}^{N} (\sup f - \inf f) \operatorname{vol}(\tilde{R}_i) \\ &\leq \alpha \operatorname{vol}(A) \end{aligned}$$

□

Theorem 3.33 (Lebesgue Criterion for integrability). *$A \subseteq \mathbb{R}^n$ rectangle and $f\colon A \to \mathbb{R}$ bounded, then f is integrable over $A \iff f$ is continuous except on a set of measure zero.*

Proposition 3.34. *Let D_f be the set of discontinuities of f, we can write $D_f = \{x\colon \omega_f(x) > 0\}$.*

$\implies$ *We have that f is integrable, let*

$$H_\alpha = \{x\colon \omega_f \geq \alpha\}$$

We show that for every $\alpha > 0$ that H_α has measure zero. In fact we prove that it has content zero. Notice that $D_f = \bigcup_{k=1}^{\infty} H_{2^{-\alpha}}$, which is a countable union of sets of measure zero, and therefore has measure zero.

Fix $\alpha > 0$ and let $\varepsilon > 0$. Now f is integrable over A, then $\exists P$ a partition of A into rectangles $R_1, \cdots, R_N$ s.t. $U(P, f) - L(P, F) < \frac{\varepsilon}{2}\alpha$. We have three cases:

i) $R_i \in c_1 \iff R_i \cap H_\alpha = \varnothing$

ii) $R_i \in c_2 \iff R_i \cap H_\alpha \subseteq \partial(R_i)$

iii) $R_i \in c_3 \iff \mathring{R}_i \cap H_\alpha \neq \varnothing$.

Notice that since $A = \bigcup_{i=1}^{N} R_i$, we have

$$H_\alpha = \left(\bigcup_{R \in c_2} R \cap H_\alpha \right) \cup \left(\bigcup_{R \in c_3} R \cap H_\alpha \right) := E_2 \cup E_3$$

- *if $R \in c_2$, then $R \cap H_\alpha \subseteq \partial R_2$, but the boundary of a rectangle has measure zero. Therefore $R \cap H_\alpha$ ha measure zero for every $R \in C_2$. Therefore E_2 is a finite union of measure zero sets and hence has measure zero and compact. Then we can cover E_2 with finitely many rectangles $\hat{R}_1, \cdots, \hat{R}_k$ s.t.*

$$\sum_{i=1}^{k} \operatorname{vol}(\hat{R}_j) < \varepsilon/2$$

- *if $R \in c_3$, then $R \cap H_\alpha \neq \varnothing$, then $\exists x_0 \in \mathring{R} \cap H_\alpha$, and $\delta_0 > 0$ s.t.*

$$N_{\delta_0}(x) \subseteq \mathring{R}$$

Now since $x_0 \in H_\alpha$

$$\sup_{x_1,x_2 \in N_{\delta_0}(x_0)} |f(x_1) - f(x_2)| = \mathrm{osc}_f(N_{\delta_0}(x_0)) \geq \omega_f(x_0) \geq \alpha$$

Therefore

$$\sup_R f - \inf_R f \geq \sup_{N_{\delta_0}(x_0)} - \inf_{N_{\delta_0}(x_0)} \geq \alpha$$

Hence

$$\frac{\varepsilon\alpha}{2} > U(P,f) - L(P,f) = \sum_i (M_i - m_i)\,\mathrm{vol}(R_i) \geq \sum_{R \in c_3} (M_r - m_r)\,\mathrm{vol}(R)$$

Hence

$$\frac{\varepsilon\alpha}{2} > \alpha \sum_{R \in C_3} \mathrm{vol}(R) \implies \sum_{R \in C_3} \mathrm{vol}(R) < \varepsilon/2$$

Therefore

$$E \subseteq_{R \in C_3} R, \quad \sum_{R \in c_3} \mathrm{vol}(R) < \varepsilon/2$$

Hence

$$H_\alpha \subset \left\{\hat{R}_1, \cdots, \hat{R}_k\right\} \cup \{R \in C_3\}$$

with total volume $< \varepsilon$. *Therefore* H_α *has measure zero.*

$\Longleftarrow$ *We have that* D_f *has measure zero, we show that* $f \in R(A)$. *Let* $\varepsilon > 0$, *we want to find* P *s.t.*

$$U(P,f) - L(P,f) < \varepsilon$$

We write $M = \sup|f|$ *and let* $\alpha = \frac{\varepsilon}{2M + \mathrm{vol}(A)}$. *Take*

$$H_\alpha = \{x \in A \colon \omega_f \geq \alpha\} \subseteq D_f$$

and D_f *has measure zero, therefore* H_α *has measure zero. Moreover,* H_α *is compact so it has content zero i.e. there exists finitely many rectangles* $R_1, \cdots, R_N$ *covering* H_α *s.t.*

$$\sum_{i=1}^{N} \mathrm{vol}(R_i) < \alpha$$

Now use the vertices of R_i *and* A *to construct a partition* P *of* A *into rectangles* $S_1, \cdots, S_L$. *We have the following cases:*

i) $S_j \in c_1 \iff S_j \cap H_\alpha \neq \varnothing$ *here* $S_j \subseteq R_i$ *for some* i.

ii) $S_j \in c_2 \iff S_j \cap H_\alpha = \varnothing$.

For $S_j \in C_2$, $S_j \cap H_\alpha = \varnothing$ *hence for* $x \in S_j$ *we have that*

$$\omega_f(x) < \alpha$$

hence by Lemma above, we can find some partition of S_j into sub-rectangles $S_{j,k}$ s.t.

$$\sum_{k=1}^{k_j}(M_{s_j,k} - m_{s_j,k}) \leq \alpha \operatorname{vol}(S_j)$$

Now use the vertices of $S_{j,k},\ S_j \in C_2$ to construct a refinement P^* of P into sub-rectangles $Q_1, \cdots, Q_T$. We now define cases for Q_i:

- *$Q_t \in c_3 \iff Q_t \subseteq S_j$ where $S_j \cap H_\alpha \neq \varnothing$. then*

$$\begin{aligned}\sum_{Q_t \in c_3}(M_{Q_t} - m_{Q_t}) \operatorname{vol}(Q_t) &\leq \sum_{Q_t \in c_3} 2M \operatorname{vol}(Q_t) \\ &= 2M \sum_{Q_t \in c_3} \operatorname{vol}(Q_t) \\ &\leq 2M \sum_{i=1}^{N} \operatorname{vol}(R_i) \\ &< 2M\alpha\end{aligned}$$

- *$Q_t \in c_4 \iff Q_t \subseteq S_j$ where $S_j \cap H_\alpha \neq \varnothing$. In this case $Q_t \subseteq S_{j,k}$ for some j, k. Hence*

$$\begin{aligned}\sum_{Q_t \in c_4}(M_{Q_t} - m_{Q_t}) \operatorname{vol}(Q_t) &= \sum_{S_j \in c_2} \sum_{Q_t \in S_{j,k}} (M_{Q_t} - m_{Q_t}) \operatorname{vol}(Q_t) \\ &\leq \sum_{S_j \in c_2} \sum_{k=1}^{k_j} (M_{S_{j_k}} - m_{S_{j_k}}) \operatorname{vol}(S_{j_k})\end{aligned}$$

This is because fixing j, $\{Q_t \colon Q_t \subseteq S_{j_k}\}$ are a refinement of the partition $\{S_{j_k} \colon k = 1, \cdots, k_j\}$ of S_j. Hence

$$\begin{aligned}\sum_{Q_t \in c_4}(M_{Q_t} - m_{Q_t}) \operatorname{vol}(Q_t) &\leq \sum_{S_j \in c_2} \alpha \operatorname{vol}(S_j) \\ &\leq \alpha \operatorname{vol}(A)\end{aligned}$$

Therefore

$$\begin{aligned}U(P^*, f) - L(P^*, f) &= \sum_{Q}(M_Q - m_Q) \operatorname{vol}(Q) \\ &= \sum_{Q \in c_3}(M_Q - m_Q) \operatorname{vol}(Q) + \sum_{Q \in c_4}(M_Q - m_Q) \operatorname{vol}(Q) \\ &< 2M\alpha + \alpha \operatorname{vol}(A) \\ &= (2M + \operatorname{vol}(A))\alpha \\ &= \varepsilon\end{aligned}$$

Therefore f is integrable.

Example 3.35. The dirichlet function is discontinuous on $\mathbb{R}$, therefore f is not integrable.

For $x \in [0,1] \times [0,1]$ and $f(x) = \begin{cases} 1 & x \neq 1/2 \\ 0 & x = 1/2,\ y \in \mathbb{Q} \\ -1 & x = 1/2,\ y \notin \mathbb{Q} \end{cases}$ Draw the picture. Note that x is discontinuous on $\{1/2\} \times [0,1]$ which has measure zero. Then f is integrable. Lte's compute the integral. Indeed, by the homework, we have that if $f, g \in \mathcal{R}(A)$ and $f = g$ except on a set of measure zero, then

$$\int_A f = \int_A g$$

Apply this here $g(x) = 1$, we get

$$\int_A f = \int_A g = 1$$

Corollary 3.36. *Given a rectangle $A \subseteq \mathbb{R}^n$ and $f\colon A \to \mathbb{R}$ is integrable and $\psi\colon (a,b) \to \mathbb{R}$ with $(a,b) \supset f(A)$ continuous, then $\psi \circ f \in \mathcal{R}(A)$.*

Proof. Let $g = \psi \circ f$, f is integrable, then Df has measure zero. Now if f is continuous at x_0, then g must be continuous at x_0. Therefore $Dg \subseteq Df$ and hence Dg has measure zero. So g is integrable. □

Consequence. If $f, g \in \mathcal{R}(A)$, then $fg \in \mathcal{R}(A)$. This is because

$$fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$$

Which is a composition of a continuous and a differentiable function.

Remark 3.37. if $f, g \in \mathcal{R}(A)$ this doesn't imply $f \circ g \in \mathcal{R}(A)$. Counter example,

$$\psi(x) = \begin{cases} 1 & x \neq 0 \\ 0 & x = 0 \end{cases}, \quad f(x) = \begin{cases} 1/q & x = p/q \text{ irreducible, or } x = 0 \\ 0 & x \in \mathbb{Q} \end{cases}$$

In the HW, we show that f is integrable in fact $Df = \mathbb{Q}$; and ψ is integrable on $[0,1]$ since $D_\psi = \{0\}$.

$$\psi \circ f = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

Which is not integrable.

Corollary 3.38. *Let C be a bounded set in $\mathbb{R}^n$ and A rectangle in $\mathbb{R}^n$ containing C. Then*

$$\chi_C(x) = \begin{cases} 1 & x \in C \\ 0 & x \notin C \end{cases}$$

is integrable iff ∂C has measure zero iff has content zero.

Proof. $D_{\chi_C} = \partial C$. □

Definition 3.39. Let C be a bounded set of boundary measure zero in $\mathbb{R}^n$, and take $f\colon C \to \mathbb{R}$. Extend f to be zero outside C, i.e. define

$$f_C(x) = \begin{cases} f(x) & x \in C \\ 0 & x \notin C \end{cases}$$

Then we define

$$\int_C f dx := \int_A f_C(x)$$

for some A a rectangle containing C (whenever RHS is defined).

Remark 3.40. Note that the definition above is well-defined, indeed let A_1, A_2 be two rectangles containing C. Now notice that $f_C = 0$ outside $A_1 \cap A_2$. Therefore by HW,

$$f_C \in \mathcal{R}(A_1) \iff f_C \in \mathcal{R}(A_2)$$

and

$$\int_{A_1} f_C dx = \int_{A_2} f_C dx$$

so the integral is well defined.

Consequences. Let C be bounded with ∂C has measure zero

- If f_1, f_2 integrable over C, then $f_1 + \alpha f_2$ is integrable over C with

$$\int_C f_1 + \alpha f_2 dx = \int_C f_1 d + \alpha \int_C f_2 dx$$

- If f, g integrable over C, $f \leq g$ then

$$\int_C f dx \leq \int_C g dx$$

- f is integrable over C iff f is continuous on C except on a set of measure zero.

Example 3.41. Let C be the triangle in $\mathbb{R}^2$ with vertices $(0,0), (0,1)$ and $(1,1)$. Calculate

$$\int_C x d(x,y)$$

note that f is integrable since it is continuous on C. Let $A = [0,1] \times [0,1]$ and let $f_c = \begin{cases} x & x \in C \\ 0 & x \end{cases}$. Let

$$P_n = \{0, 1/n, 2/n, \cdots, 1\}$$

on $[0,1]$. Let $R_{ij} = [\frac{i-1}{n}, \frac{i}{n}] \times [\frac{j-1}{n}, \frac{j}{n}]$ and let P be the corresponding partition of $[0,1] \times [0,1]$. We have cases:

- if $j \leq i-1$: $m_{ij} = \frac{i-1}{n},\ M_{ij} = \frac{i}{n}$.
- if $j = i$: $m_{ij} = 0,\ M_{ij} = \frac{i}{n}$.
- $j = i+1$: $m_{ij} = 0, \quad M_{ij} = \frac{i}{n}$.
- $j > i+1$: $m_{ij} = 0, \quad M_{ij} = 0$.

Therefore

$$\begin{aligned}
L(P,f) &= \sum_{i,j=1}^{n} m_{i,j}\,\mathrm{vol}(R_{i,j}) \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} m_{i,j} \\
&= \frac{1}{n^2}\sum_{i=2}^{n}\sum_{j=1}^{i-1} \frac{i-1}{n}\cdot\frac{1}{n^2} \\
&= \frac{1}{n^3}\sum_{i=1}^{n}(i-1)^2 \\
&= \frac{(n-1)n(2n-1)}{6n^3}
\end{aligned}$$

$$\begin{aligned}
L(P,f) &= \sum_{i=1}^{n}\sum_{j=1}^{n} M_{i,j}\,\mathrm{vol}(R_{i,j}) \\
&= \frac{1}{n^2}\left(\sum_{i=1}^{n}\sum_{j=1}^{i}\frac{i}{n} + \sum_{i=1}^{n-1}\frac{i}{n}\cdot\frac{1}{n}\right) \\
&= \frac{1}{n^3}\left(\sum_{i=1}^{n} i^2 + \sum_{i=1}^{n-1} i\right) \\
&= \frac{n(n+1)(2n+1)+3(n-1)n}{6n^3}
\end{aligned}$$

And $f(x,y)$ is integrable with

$$L(P,f) \le \int_C x d(x,y) \le U(P,f)$$

Therefore for all n we have

$$\frac{n(n-1)(2n+1)}{6n^3} \le \int_C x d(x,y) \le \frac{n(n+1)(2n+1)+3(n-1)n}{6n^3}$$

let $n \to \infty<$ we get

$$\int_C x d(x,y) = 1/3$$

If only there was a shorter way...

3.3 Fubini's Theorem

Theorem 3.42 (Fubini's theorem). *$A \subseteq \mathbb{R}^n$ rectangle, and $f\colon A \to \mathbb{R}$ bounded. Then*

$$\underline{\int_A} f dx = \inf_P L(P,f), \quad \overline{\int_A} f dx = \sup_P U(P,f)$$

Fubini's theorem: $A \subseteq \mathbb{R}^n$ a rectangle, and $B \subseteq \mathbb{R}^m$ a rectangle. With $x \in A,\ y \in B$

$$f := f(x,y)\colon\ A \times B \to \mathbb{R}$$

and f integrable over $A \times B$, then the functions

$$\mathcal{L}(y) = \underline{\int_A} f(x,y)dx, \quad \mathcal{U}(y) = \overline{\int_A} f(x,y)dx$$

are integrable over B with

$$\int_B \left(\underline{\int_A} f(x,y)dx \right) dy = \int_B \left(\overline{\int_A} f(x,y)dx \right) = \int_{A\times B} f(x,y)d(x,y)$$

By symmetry

$$\int_{A\times B} f(x,y)d(x,y) = \int_A \left(\underline{\int_B} f(x,y)dy \right) dx = \int_A \left(\overline{\int_B} f(x,y)dy \right) dx$$

In the particular case when

$$x \to f(x,y)$$

is integrable on A for every y, we get

$$\int_{A\times B} \int_B \left(\int_A f(x,y)dx \right) dy = \int_A \left(\int_B f(x,y)dy \right) dx$$

Corollary 3.43. *By induction, if f is continuous over $[a_1,b_1] \times [a_2,b_2] \times \cdots \times [a_n,b_n]$ then*

$$\int_{[a_1,b_1]\times\cdots\times[a_n,b_n]} f(x)dx = \int_{[a_1,b_1]} \int_{[a_2,b_2]} \cdots \int_{[a_n,b_n]} f(x)dx_n \cdots dx_1$$

Example 3.44. Use Fubini to calculate $\int_C xd(x,y)$ where C is the triangle like above.